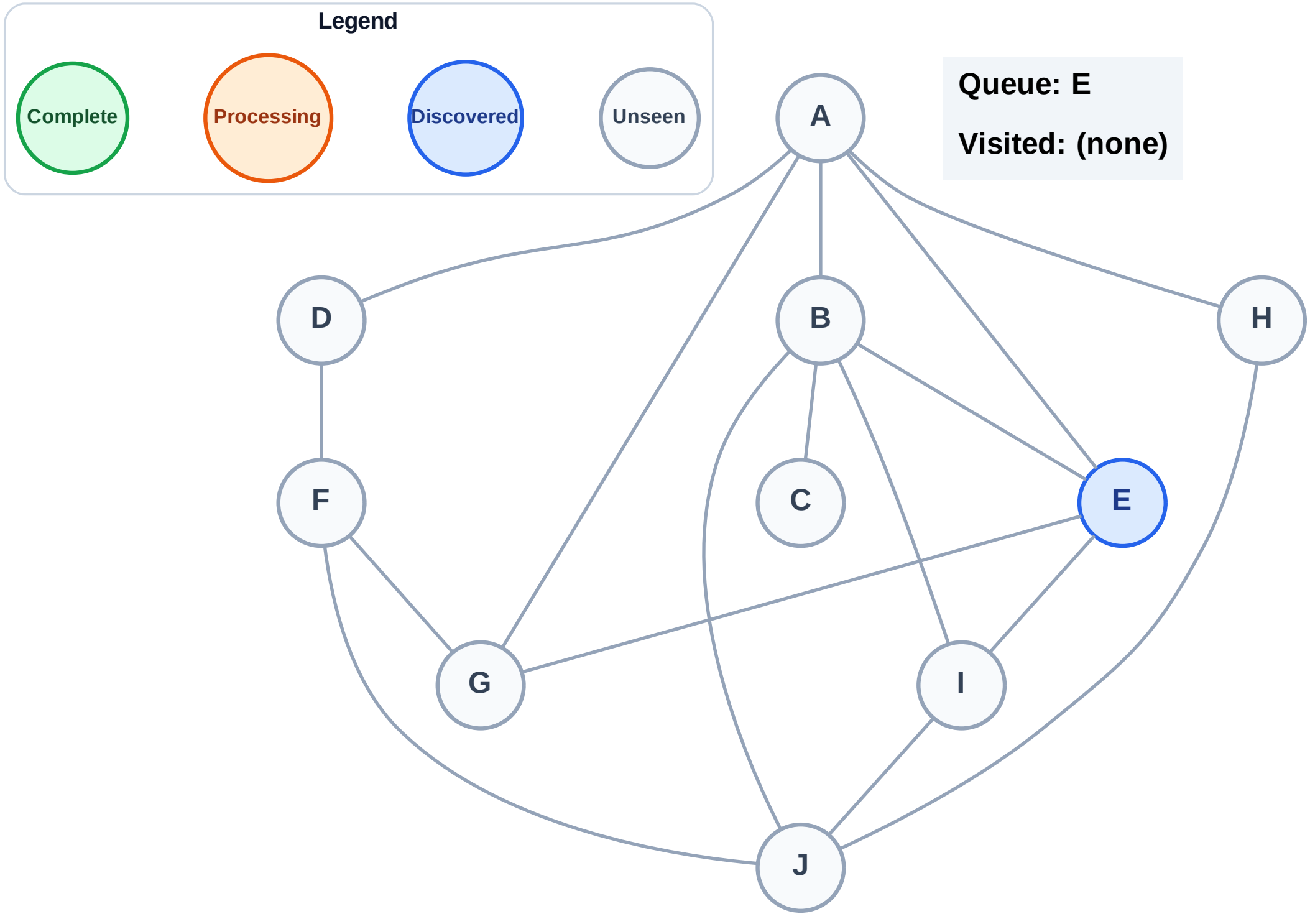
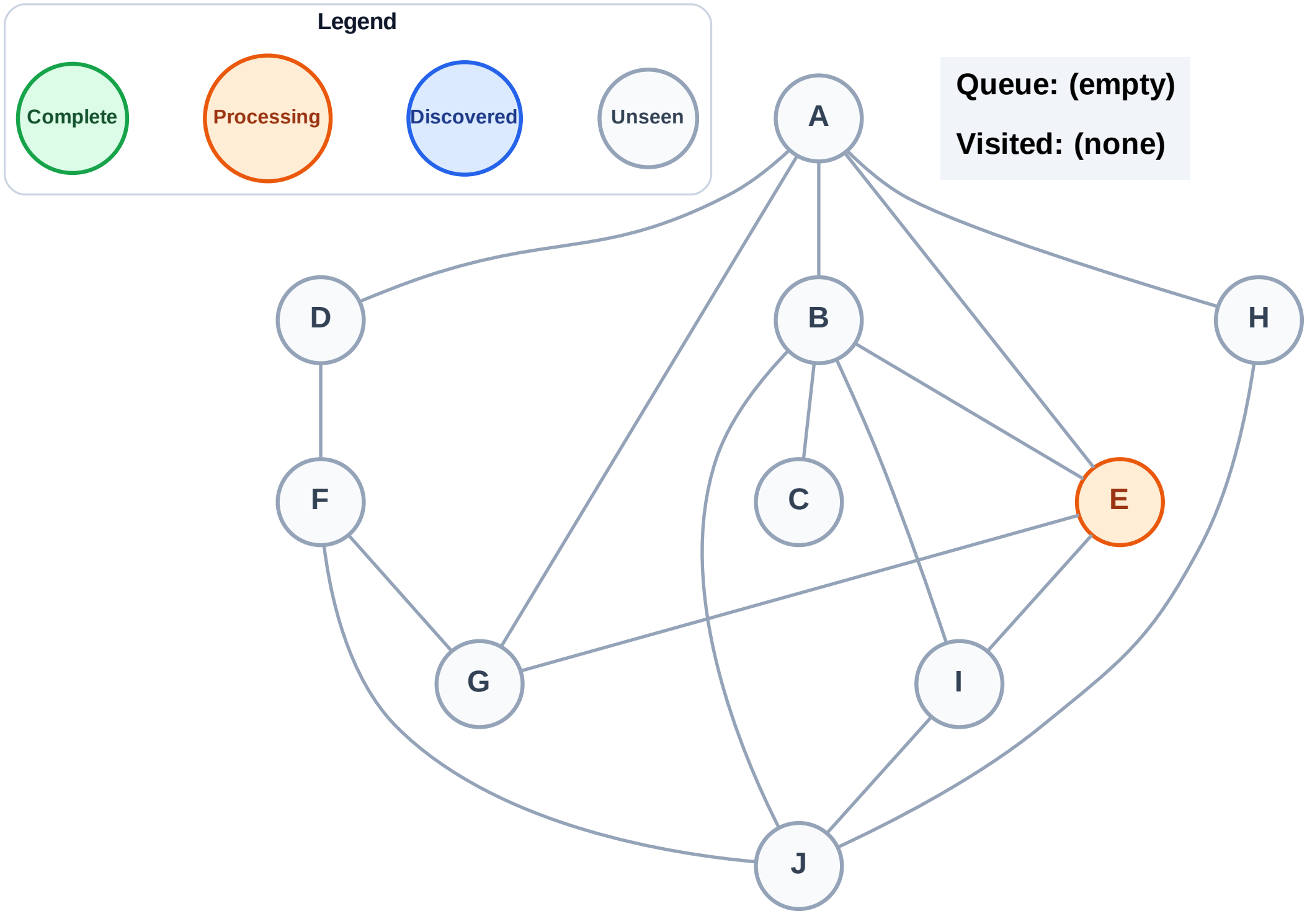
BFS Traversal

Step 1: Start at E



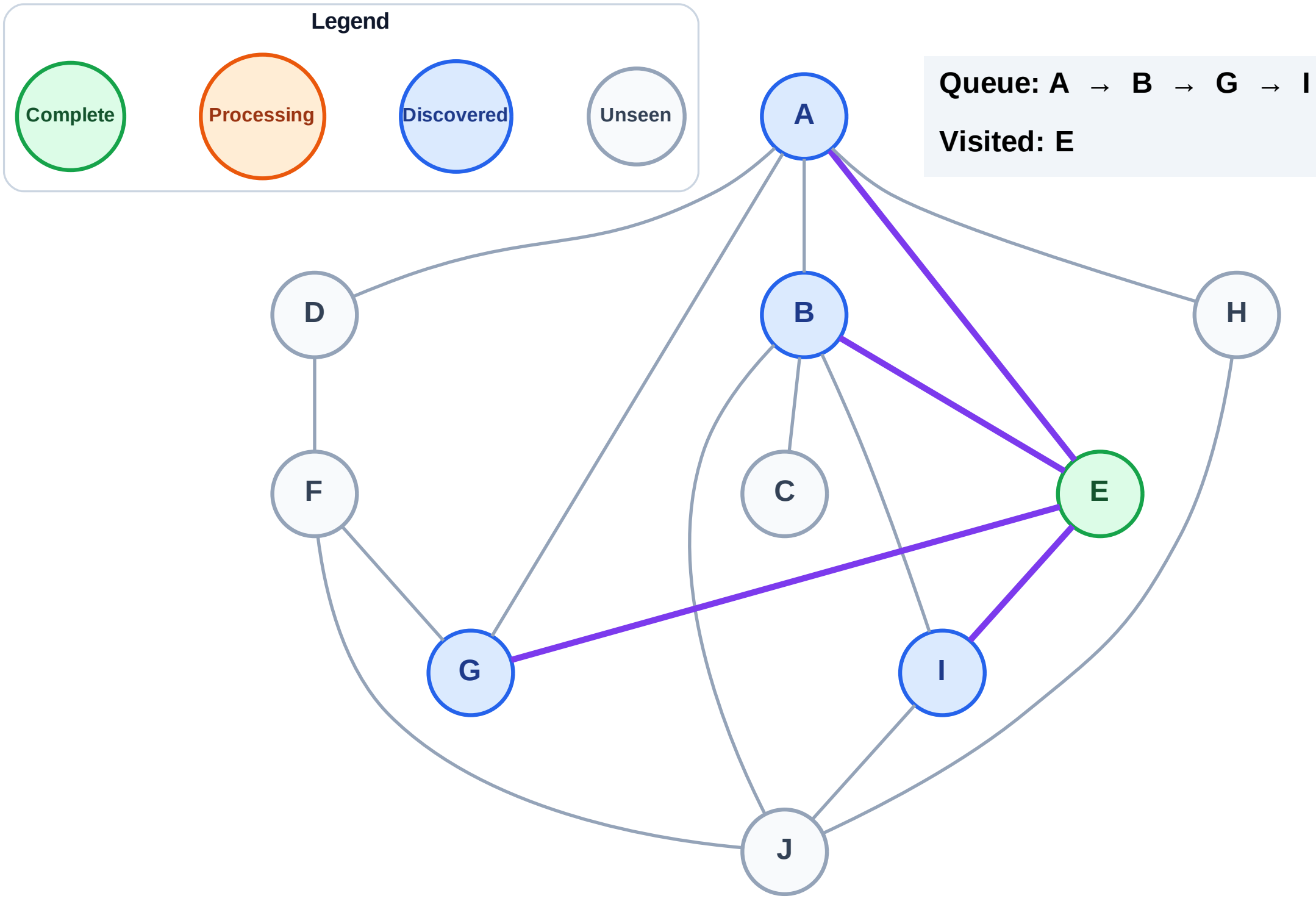
BFS Traversal

Step 2: Process node E



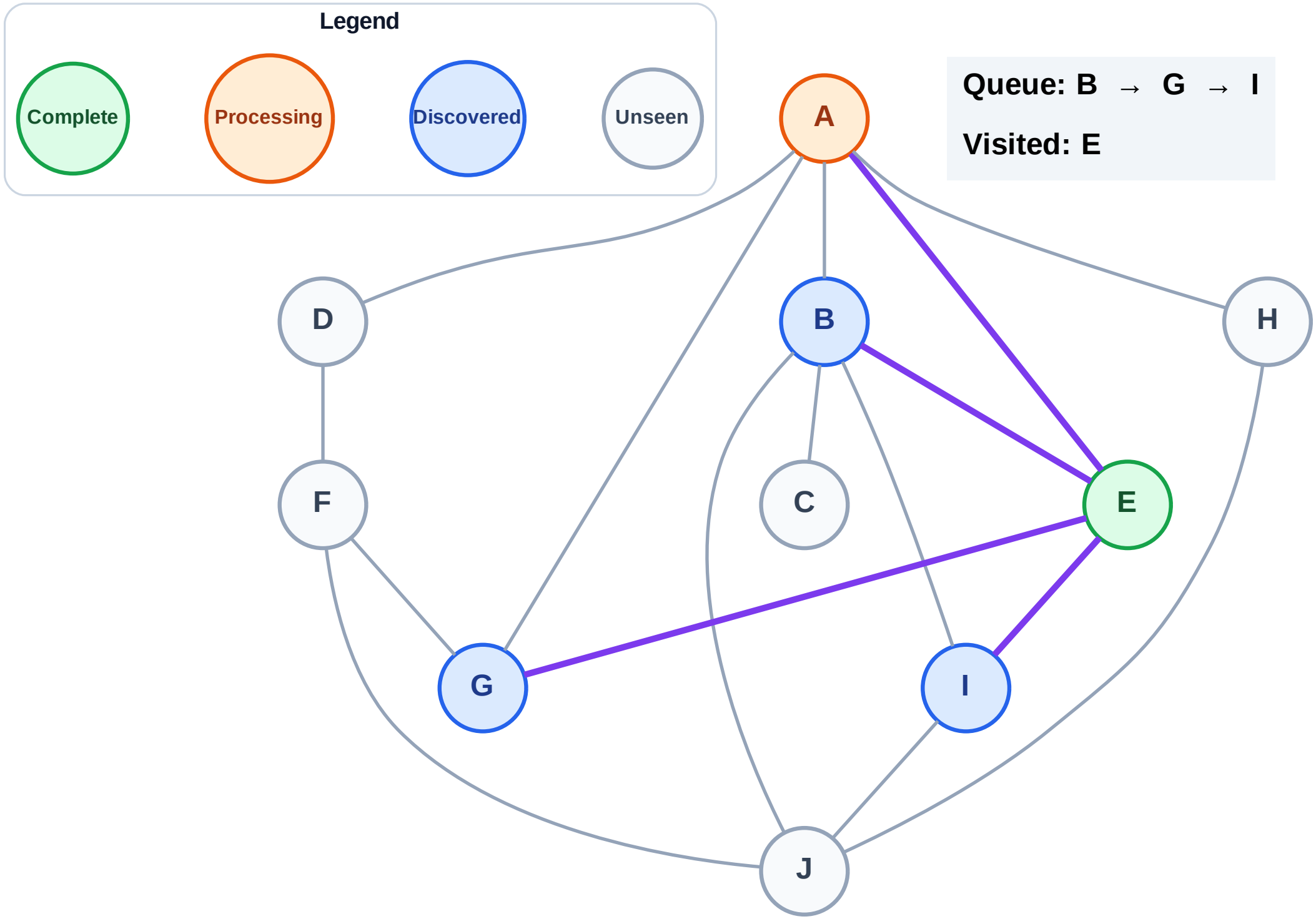
BFS Traversal

Step 3: Discover A, B, G, I from E



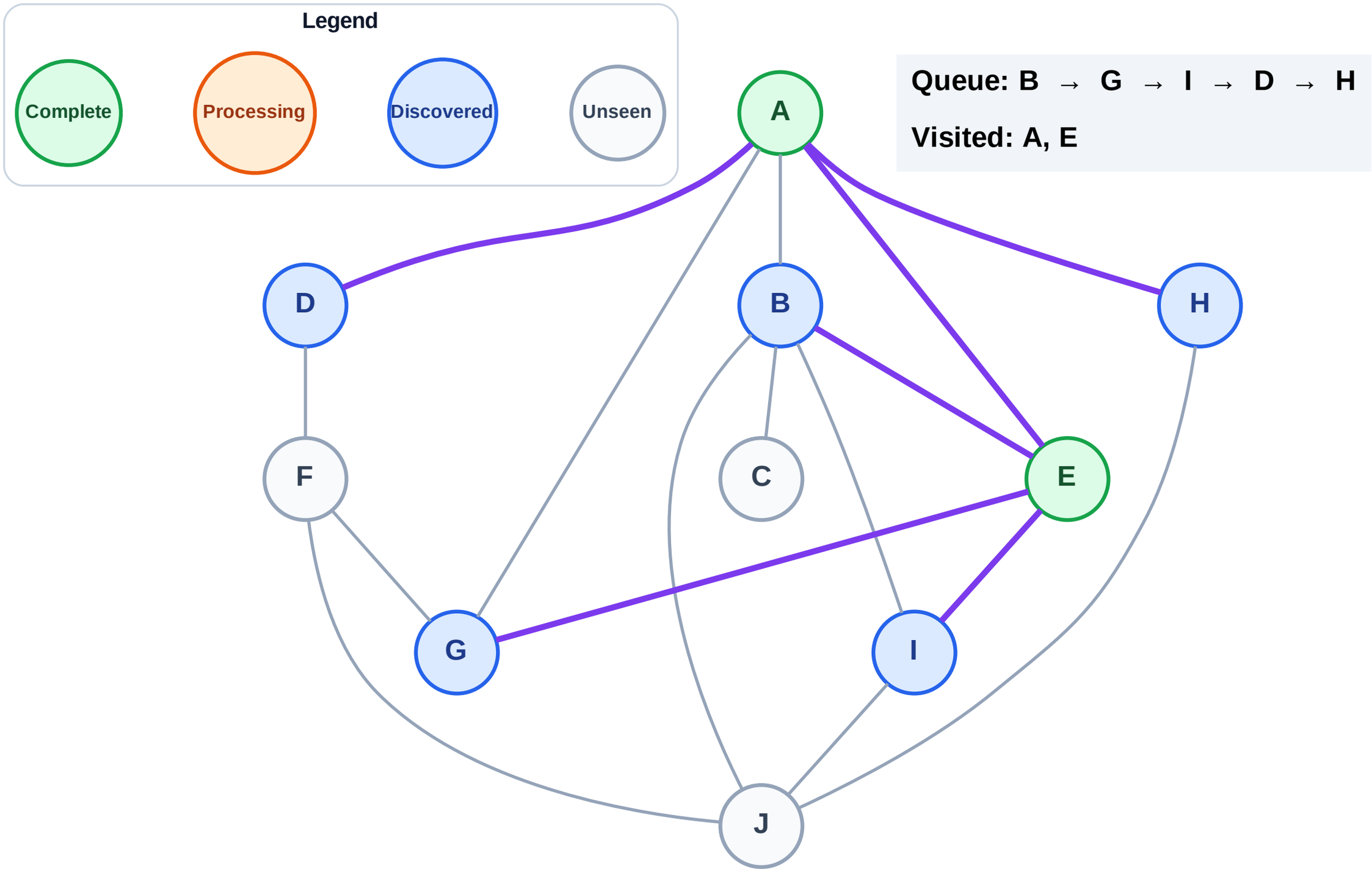
BFS Traversal

Step 4: Process node A



BFS Traversal

Step 5: Discover D, H from A

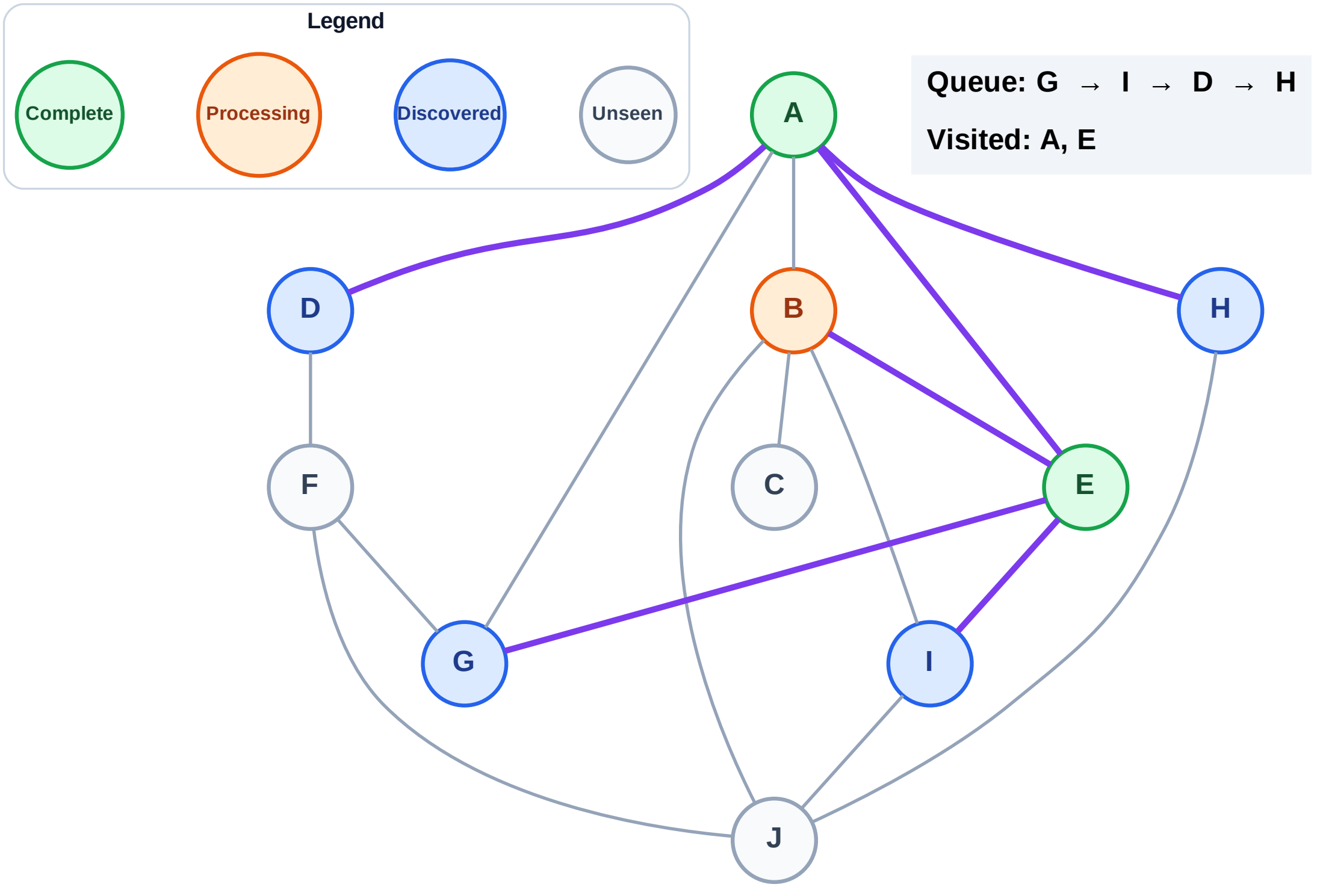


Queue: B → G → I → D → H

Visited: A, E

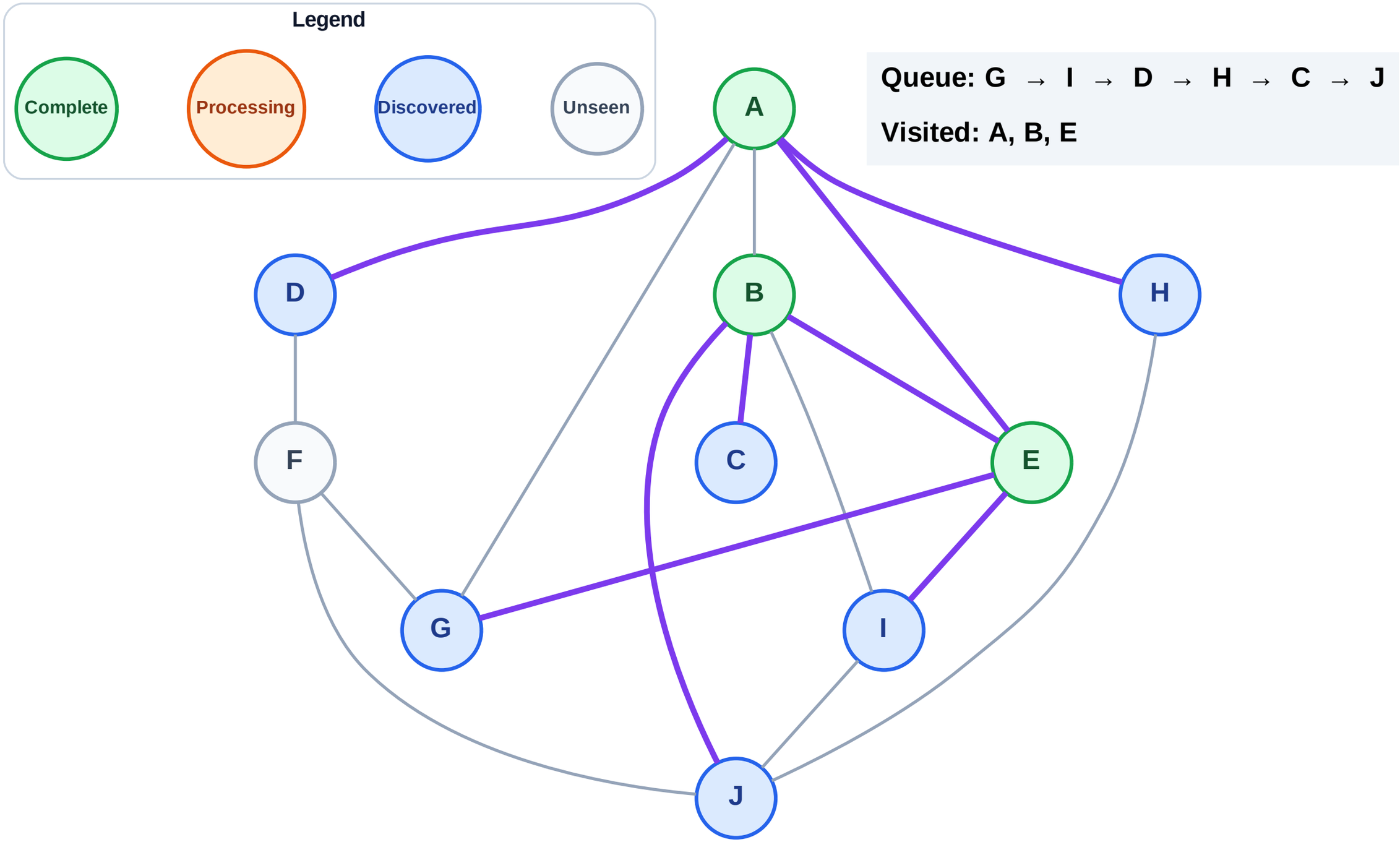
BFS Traversal

Step 6: Process node B



BFS Traversal

Step 7: Discover C, J from B

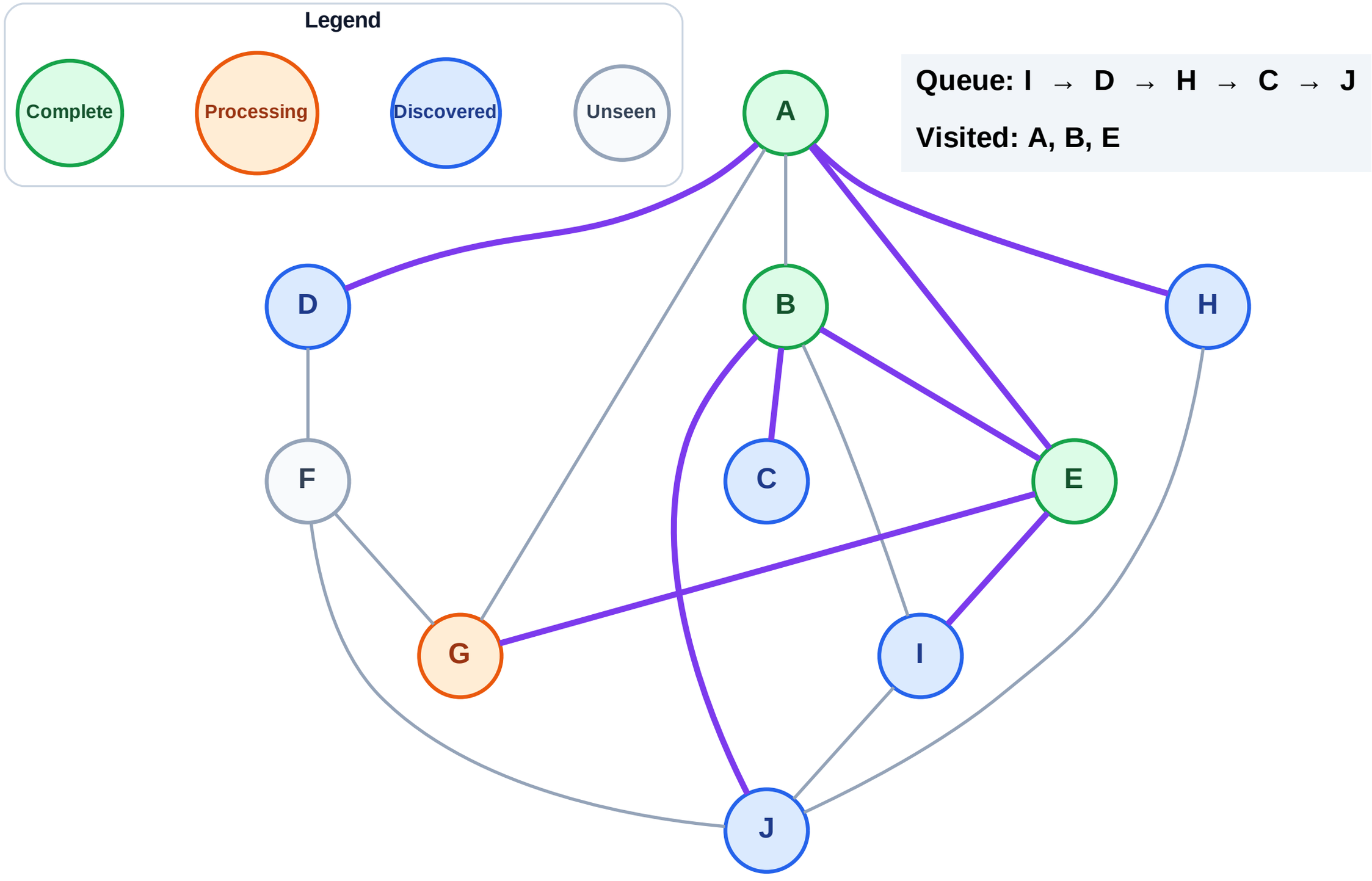


Queue: G → I → D → H → C → J

Visited: A, B, E

BFS Traversal

Step 8: Process node G

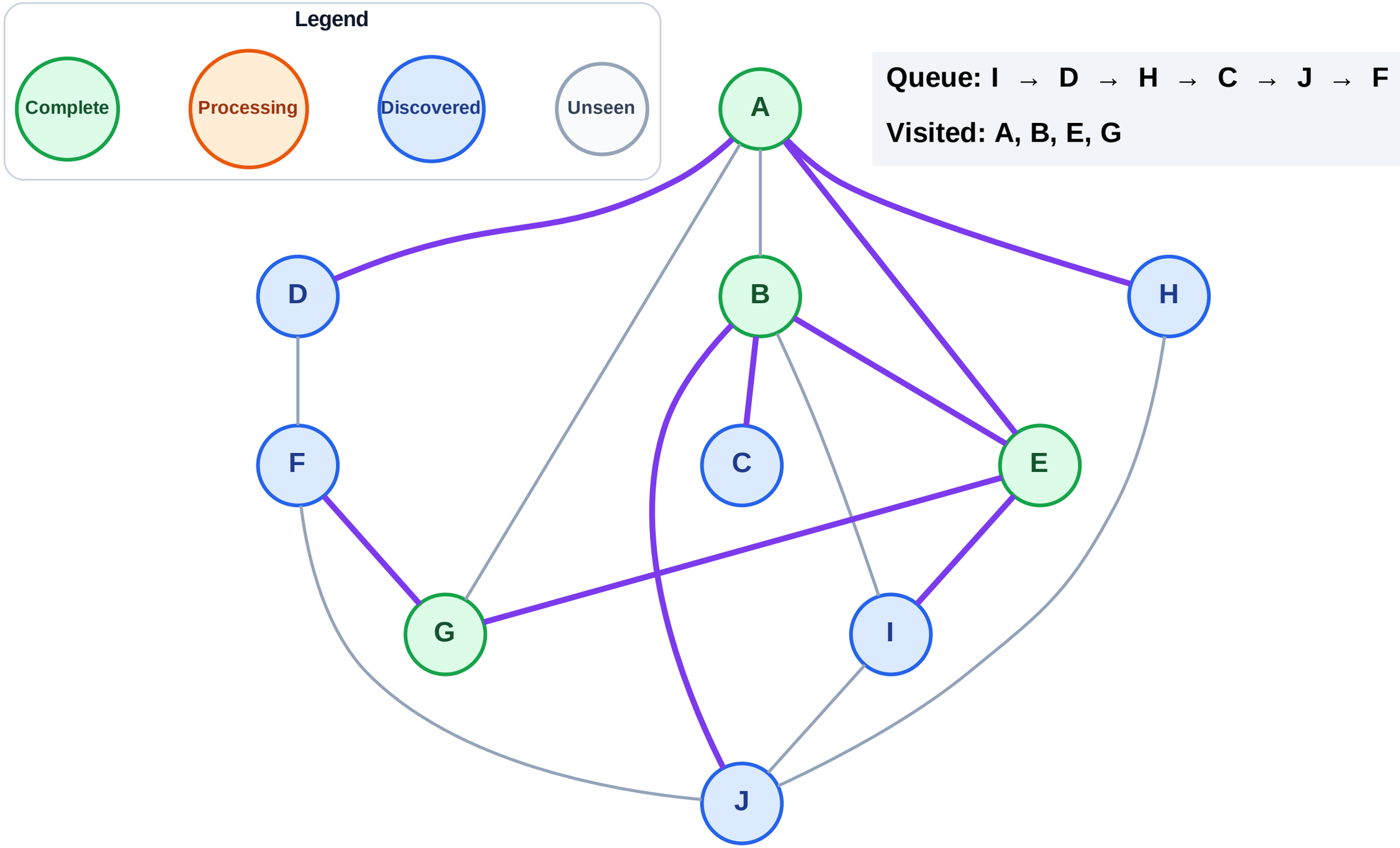


Queue: I → D → H → C → J

Visited: A, B, E

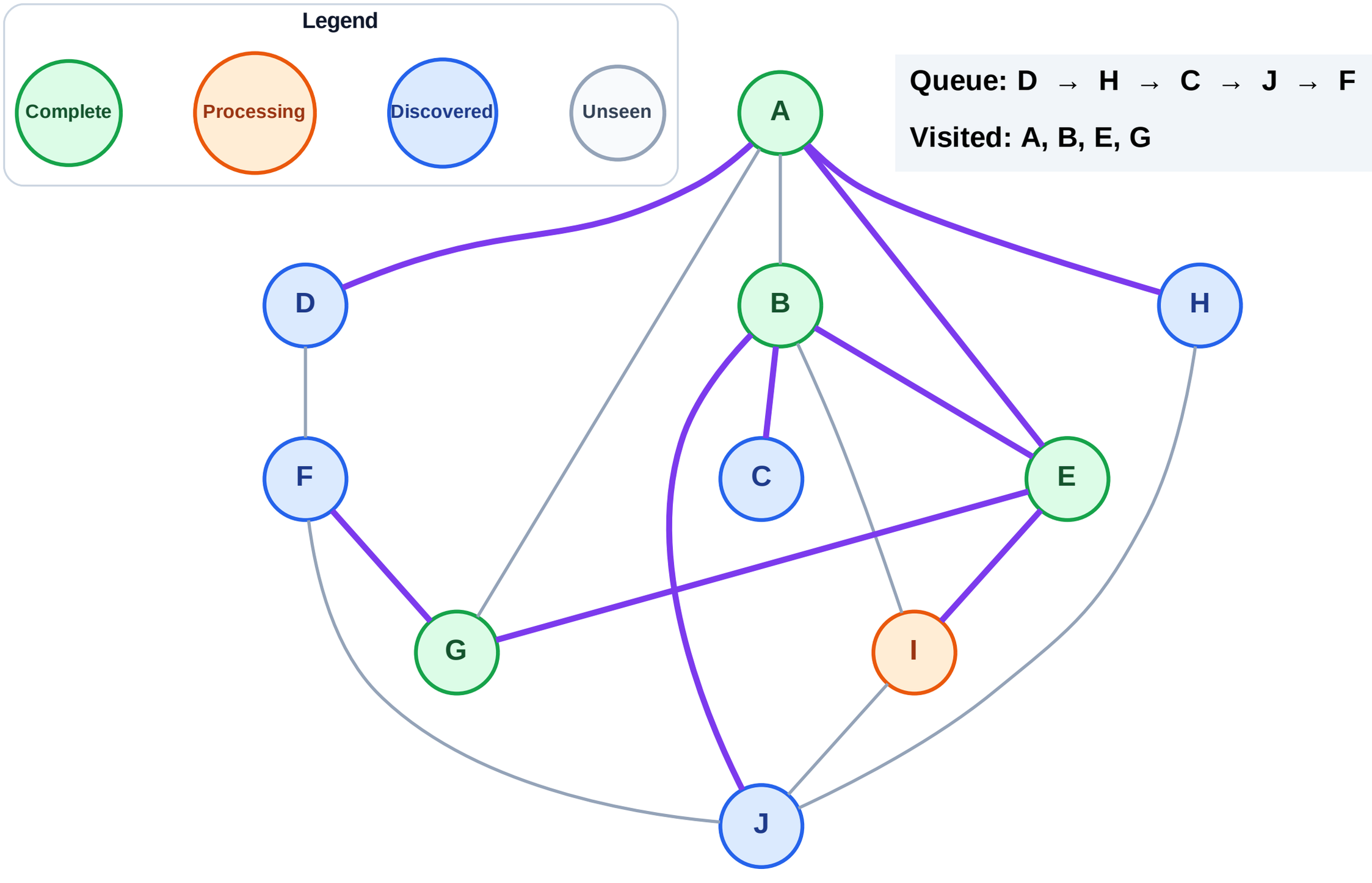
BFS Traversal

Step 9: Discover F from G



BFS Traversal

Step 10: Process node I

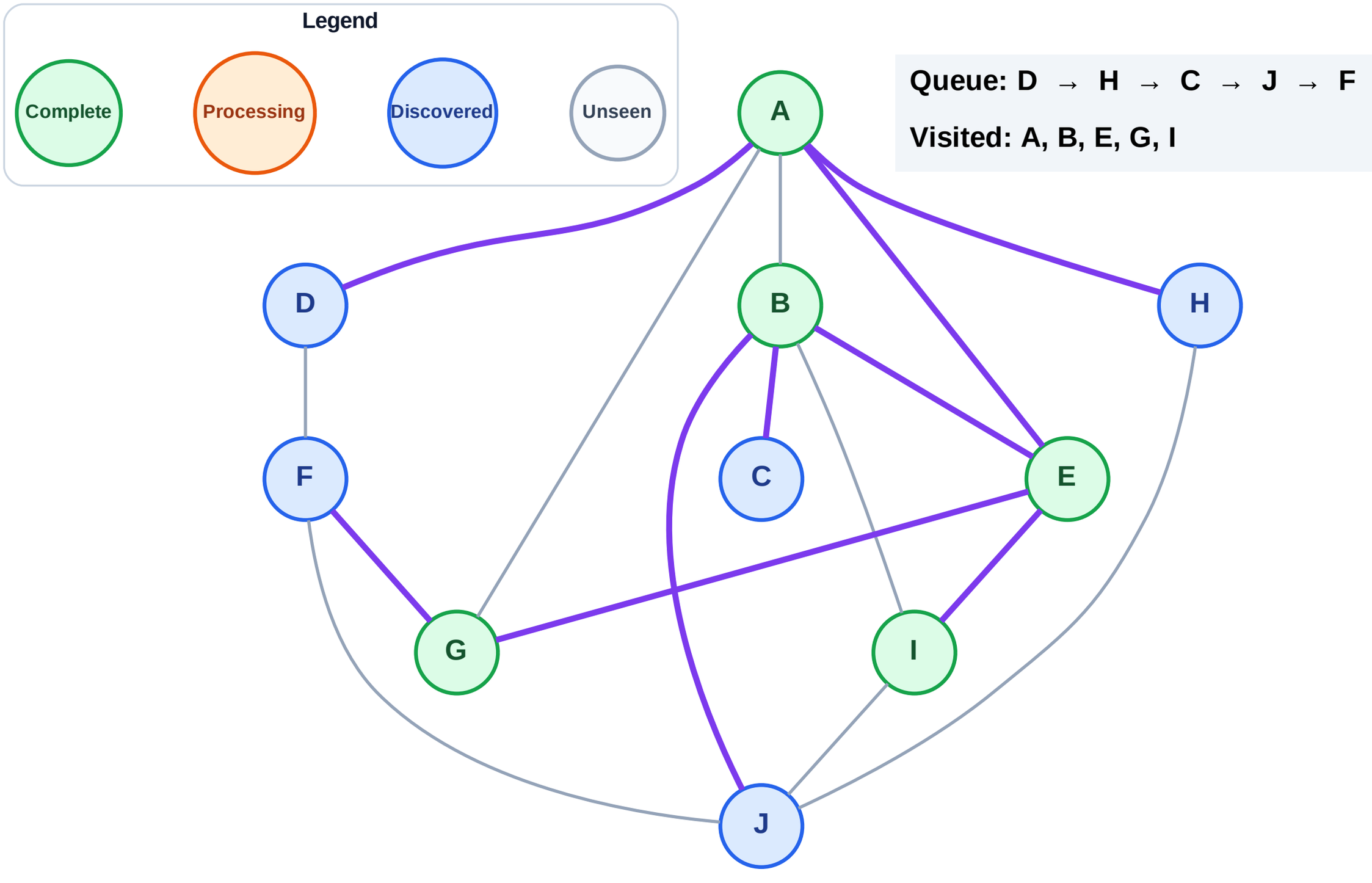


Queue: D → H → C → J → F

Visited: A, B, E, G

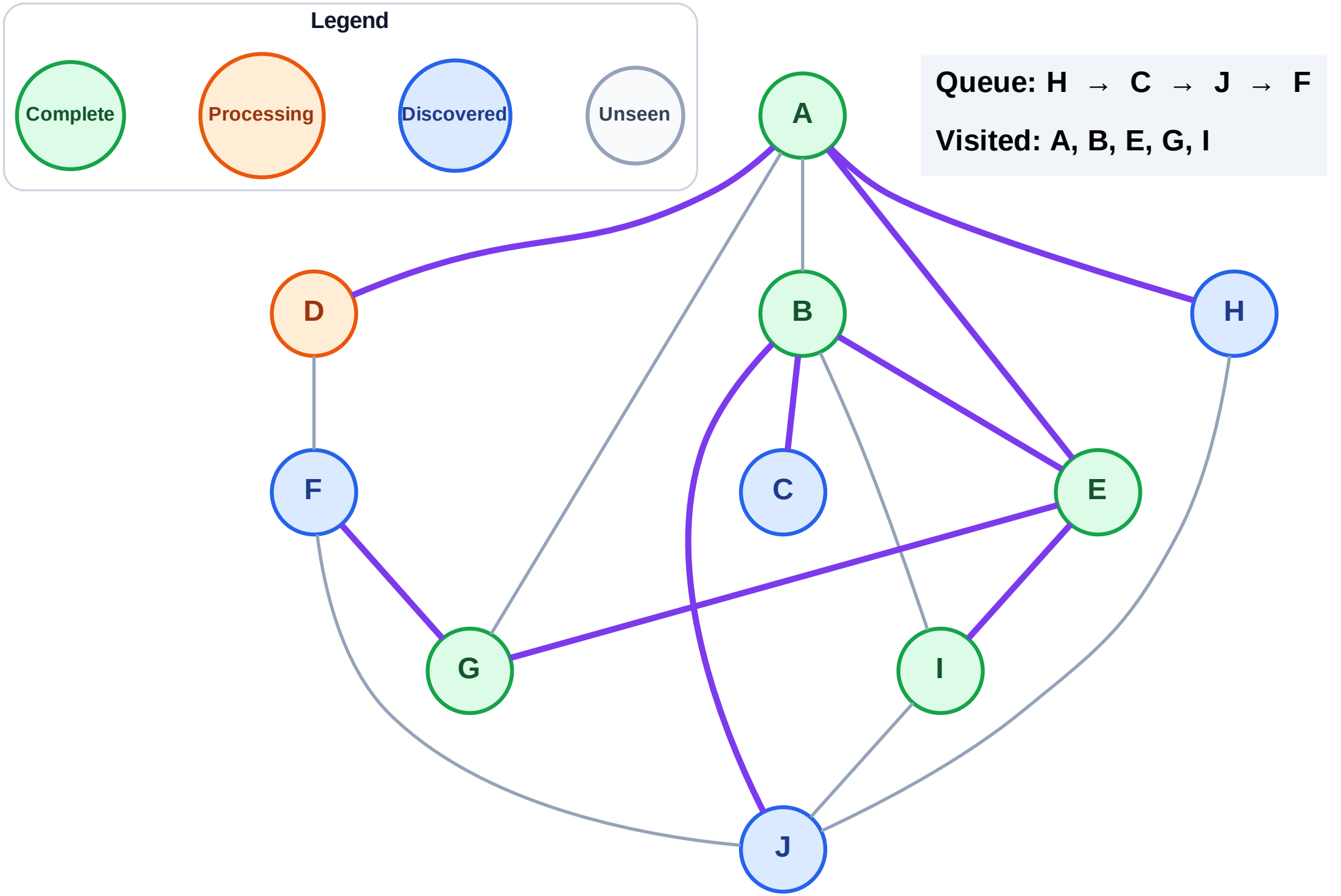
BFS Traversal

Step 11: No new neighbors from I



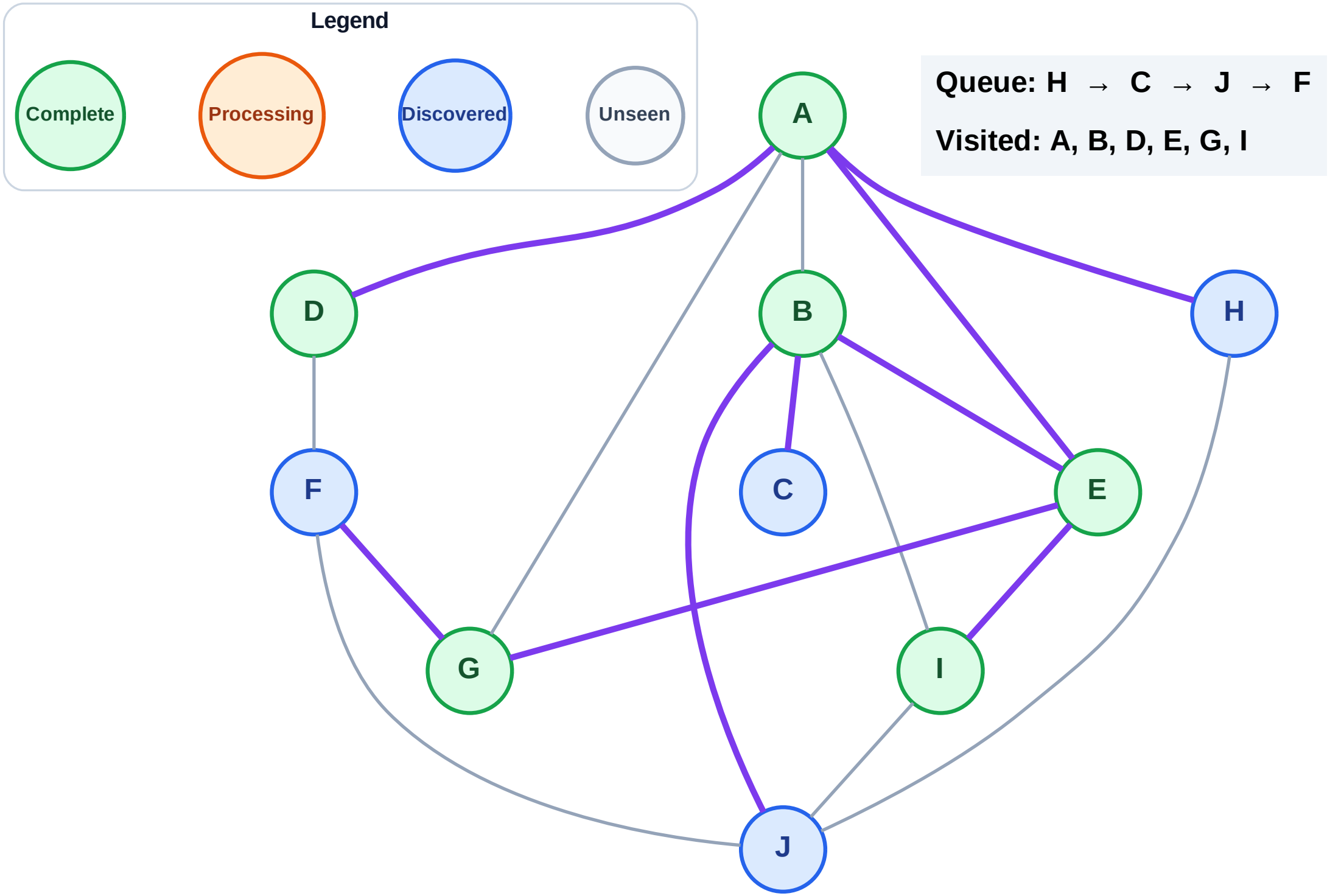
BFS Traversal

Step 12: Process node D



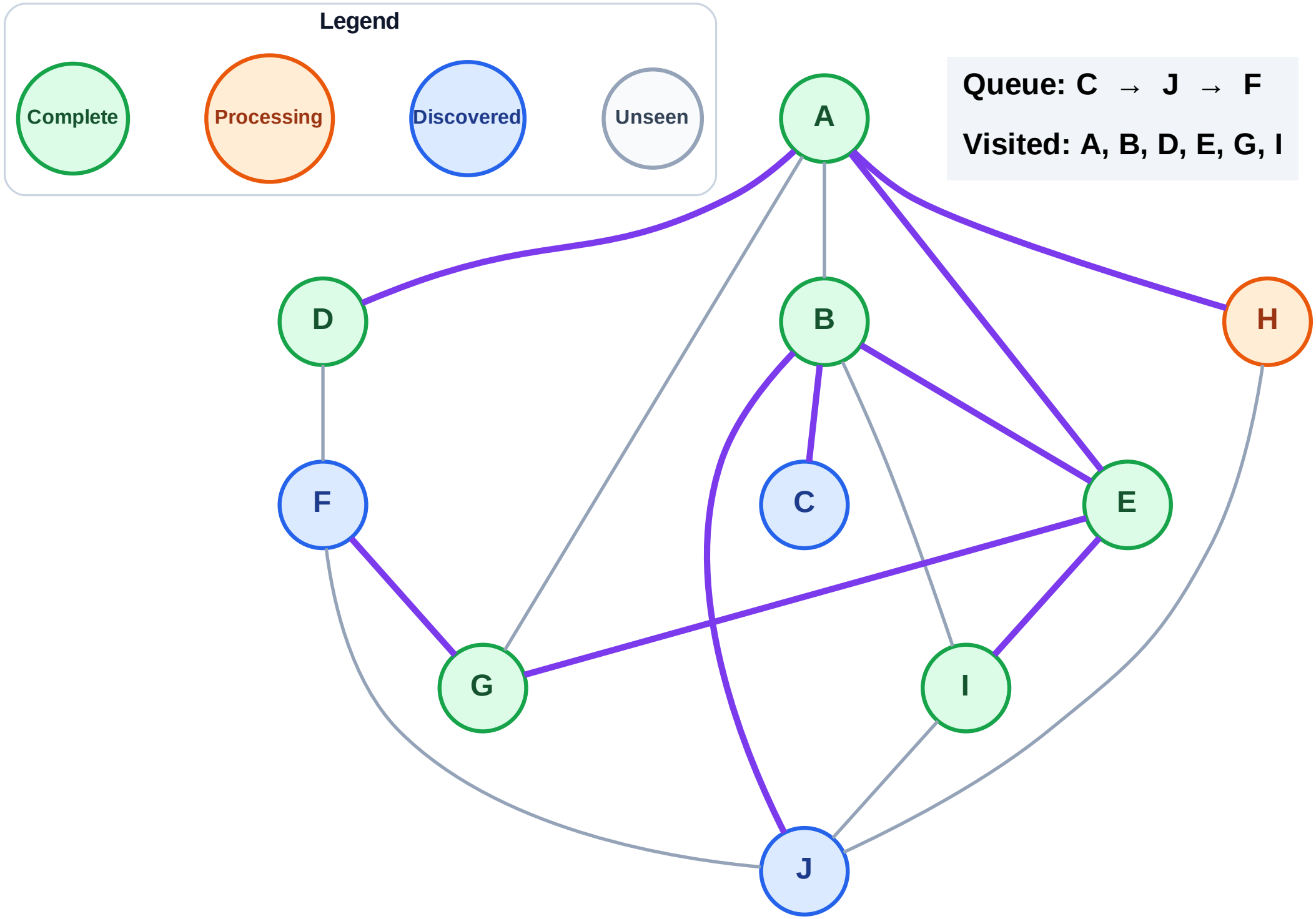
BFS Traversal

Step 13: No new neighbors from D



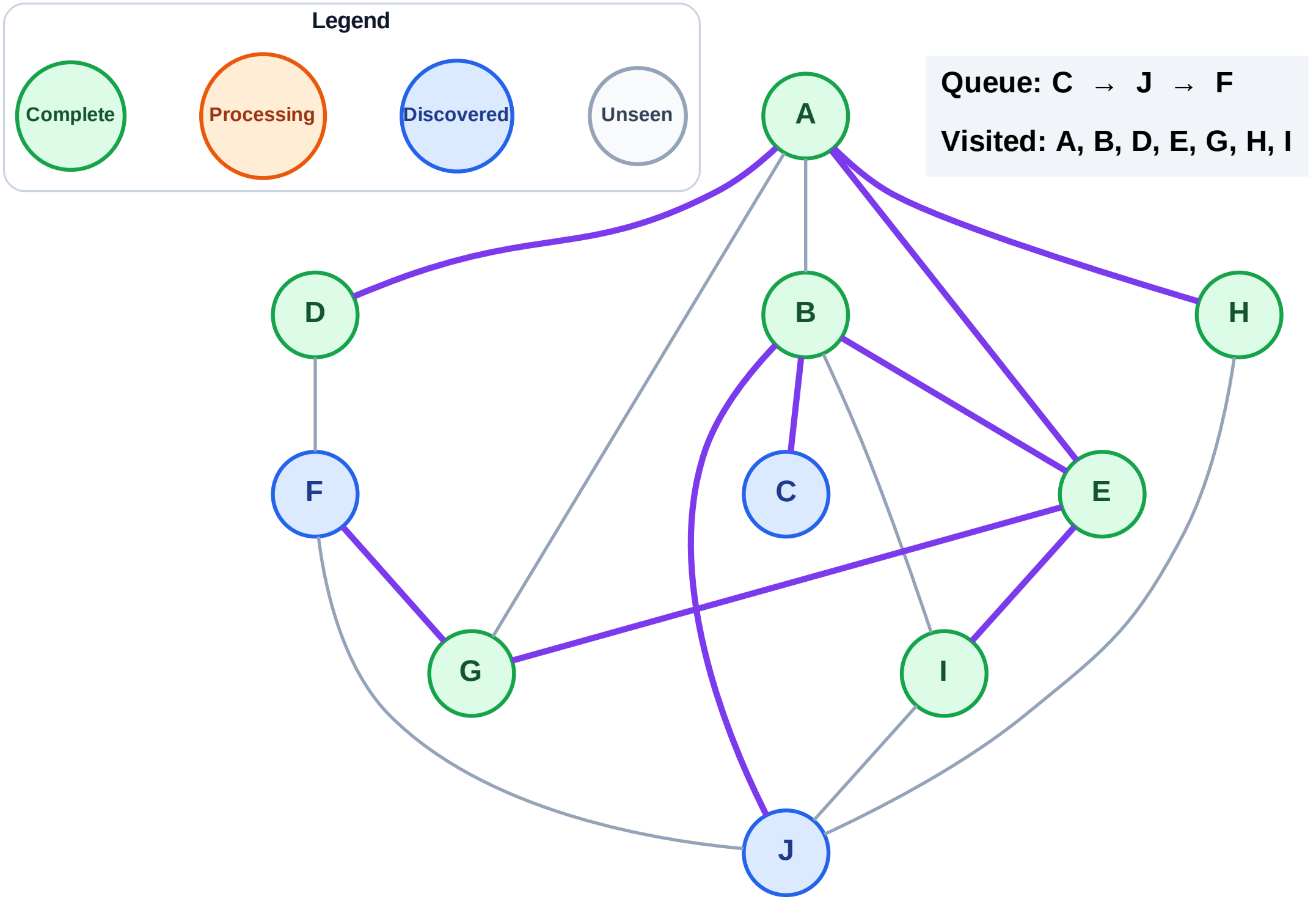
BFS Traversal

Step 14: Process node H



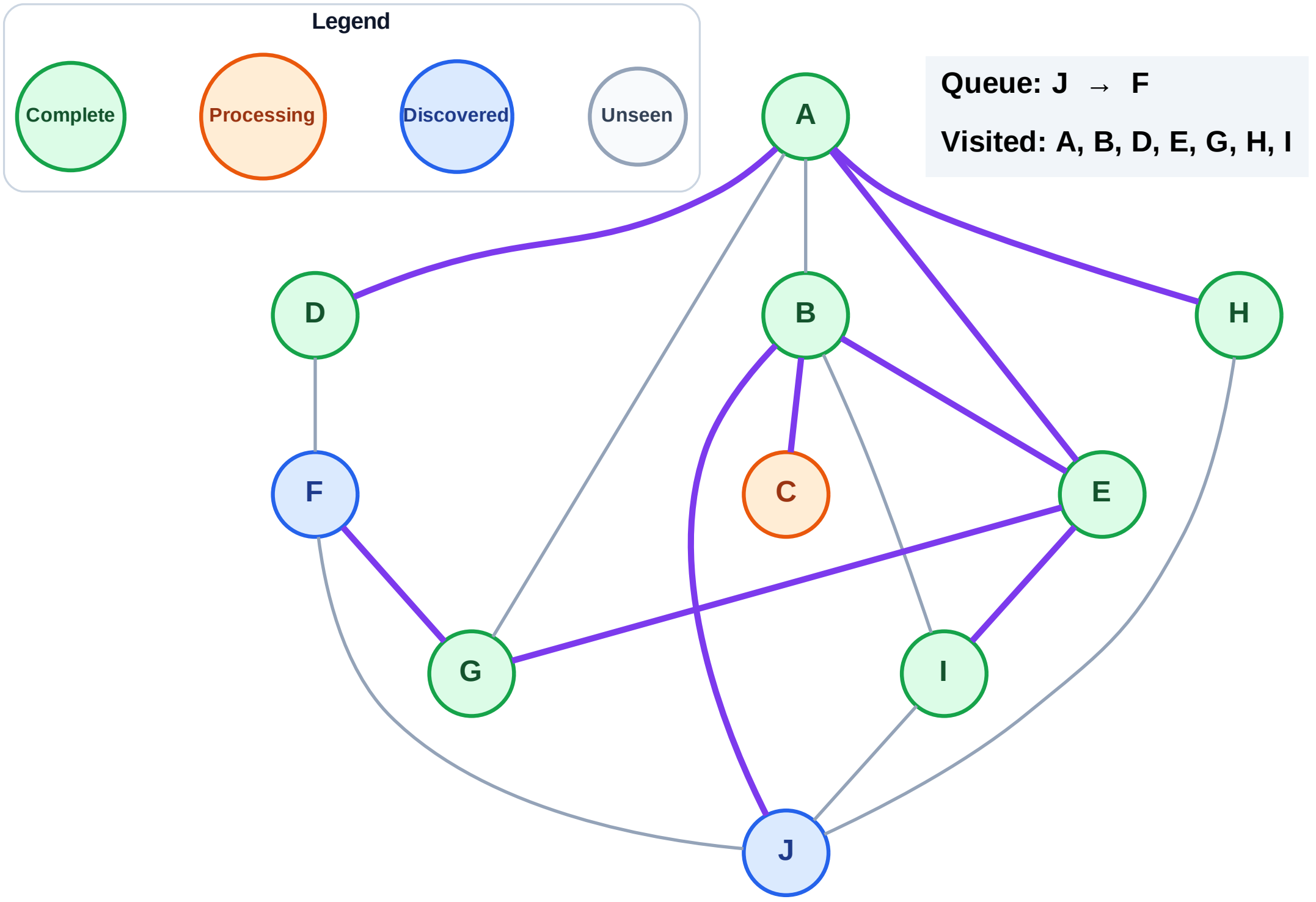
BFS Traversal

Step 15: No new neighbors from H



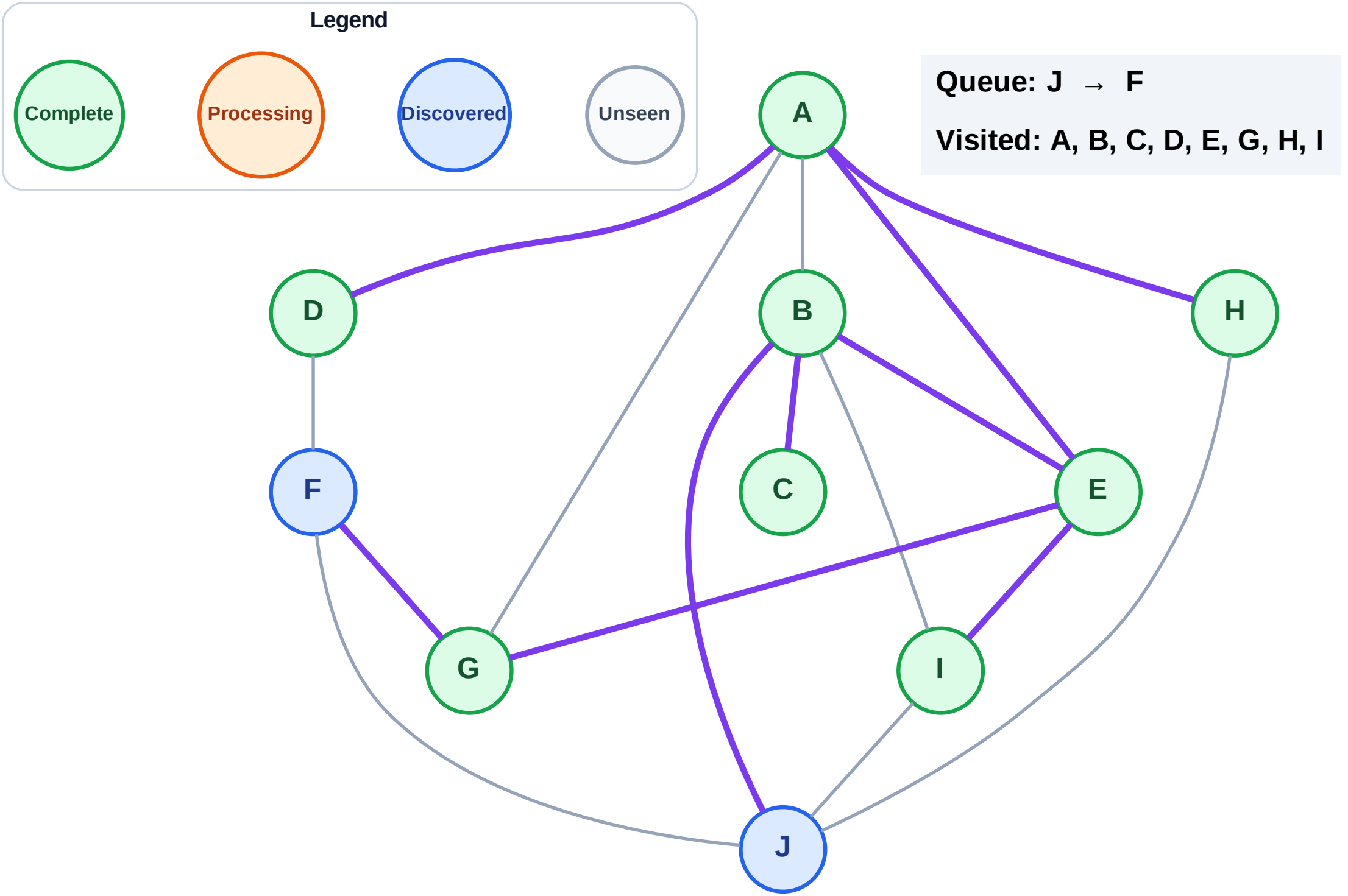
BFS Traversal

Step 16: Process node C



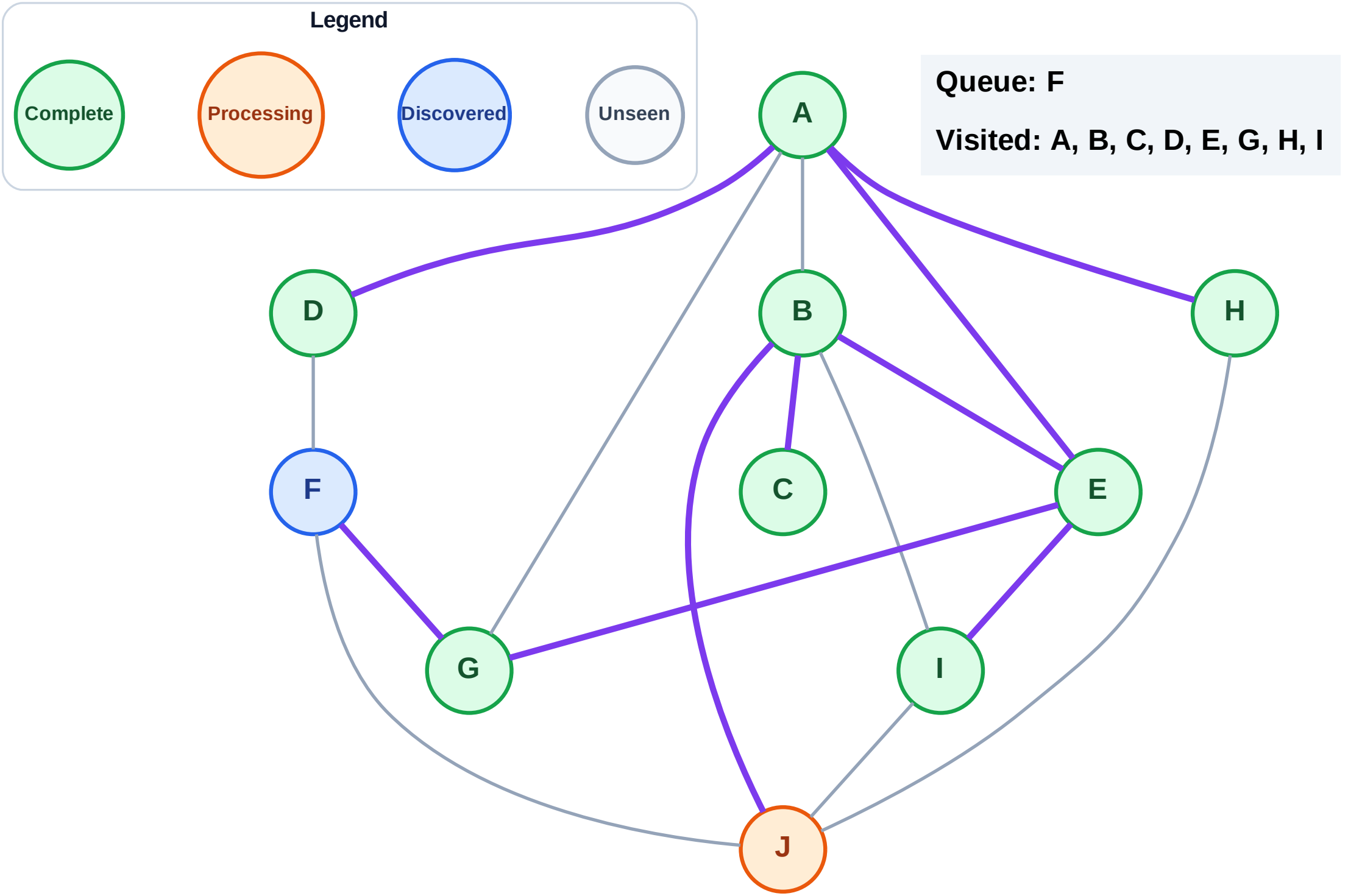
BFS Traversal

Step 17: No new neighbors from C



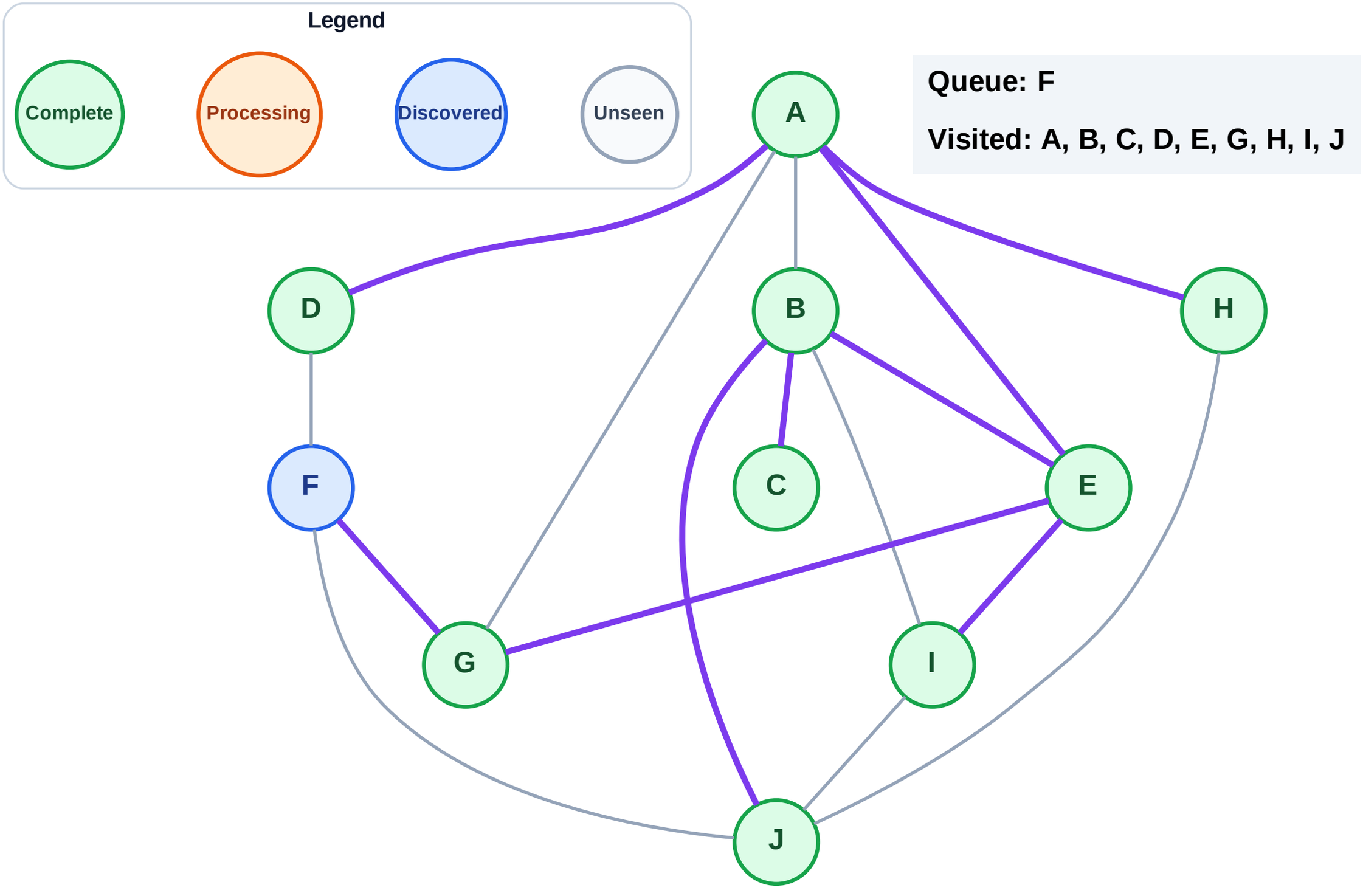
BFS Traversal

Step 18: Process node J



BFS Traversal

Step 19: No new neighbors from J

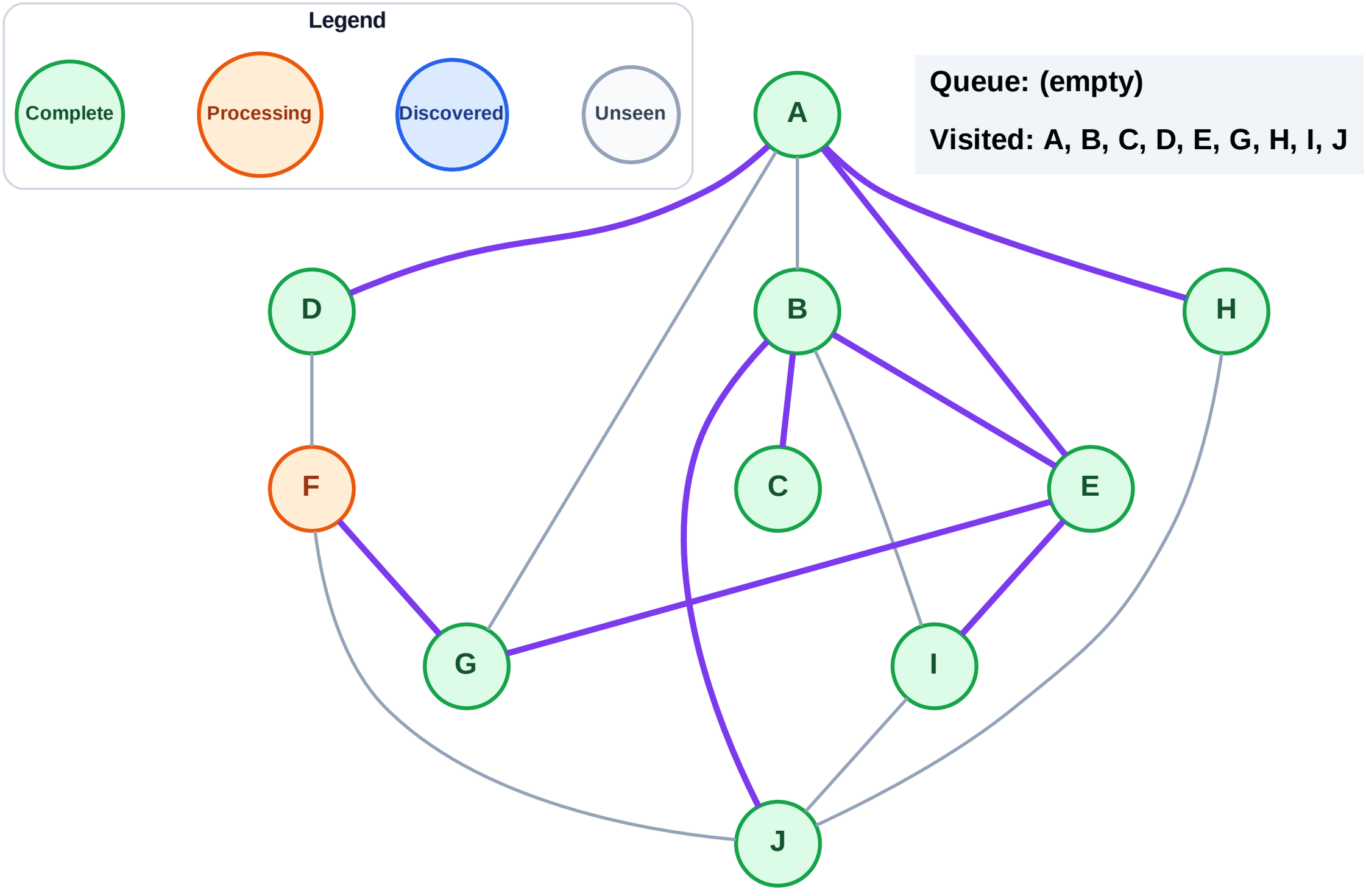


Queue: F

Visited: A, B, C, D, E, G, H, I, J

BFS Traversal

Step 20: Process node F

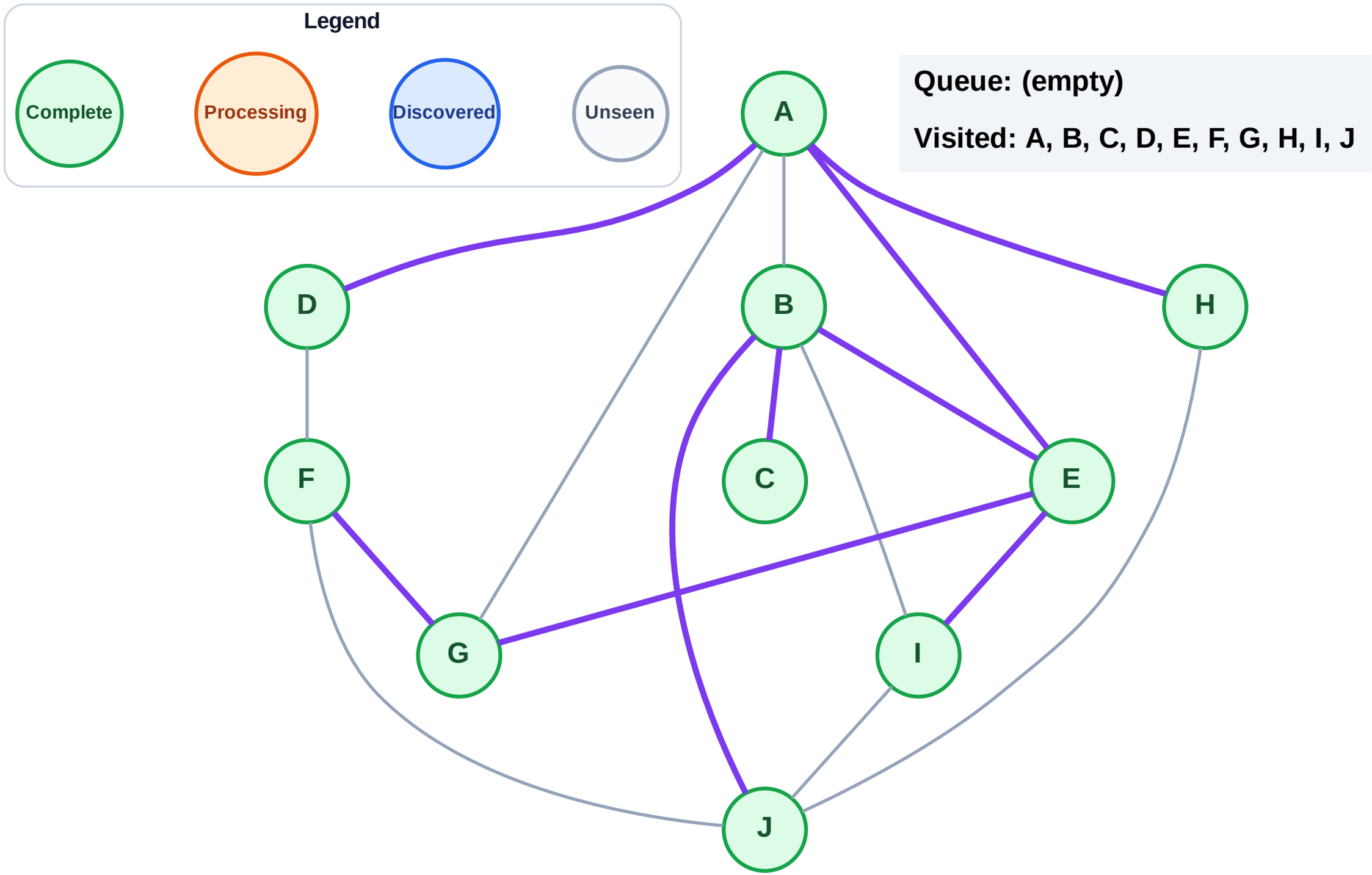


Queue: (empty)

Visited: A, B, C, D, E, G, H, I, J

BFS Traversal

Step 21: No new neighbors from F

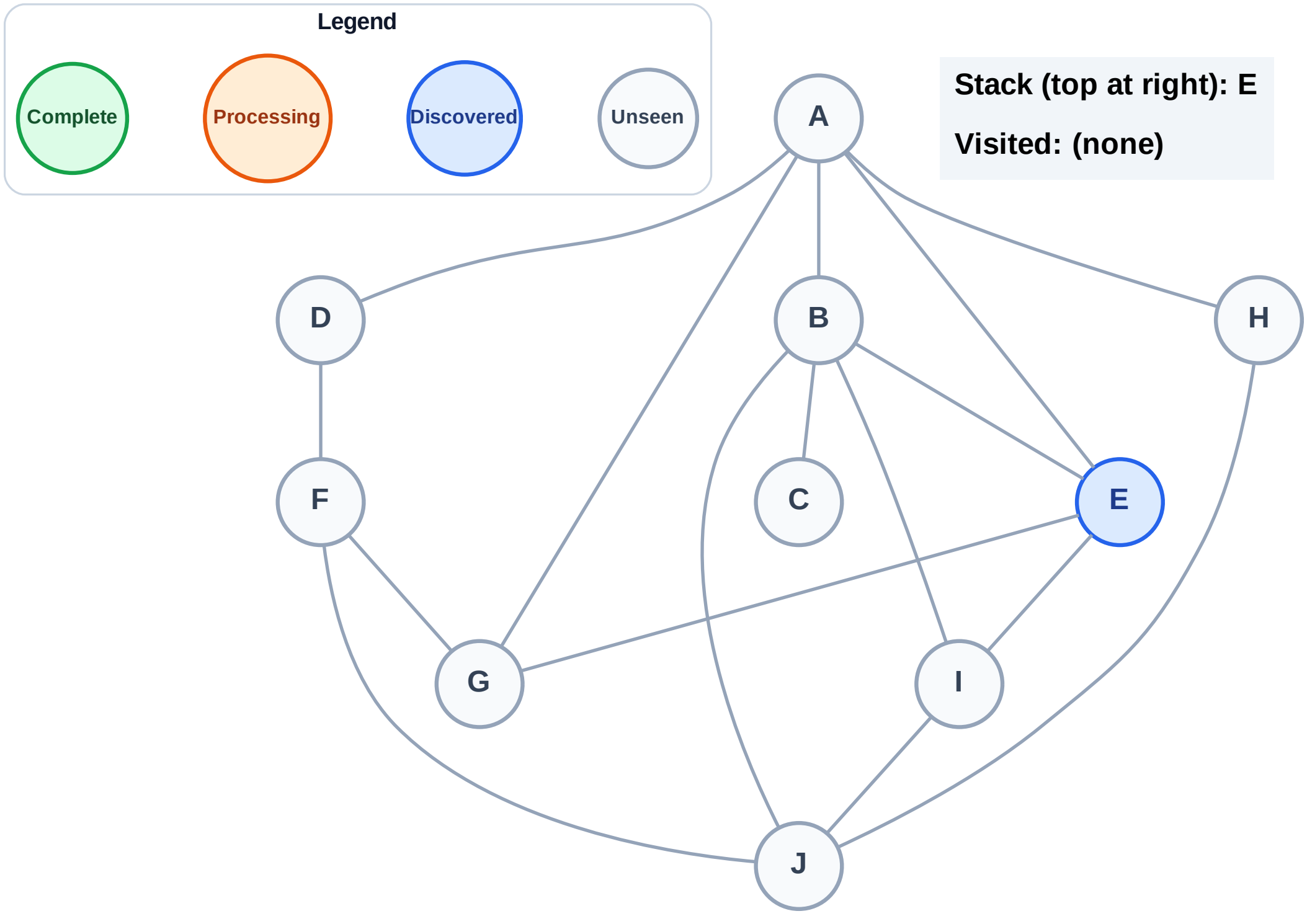


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

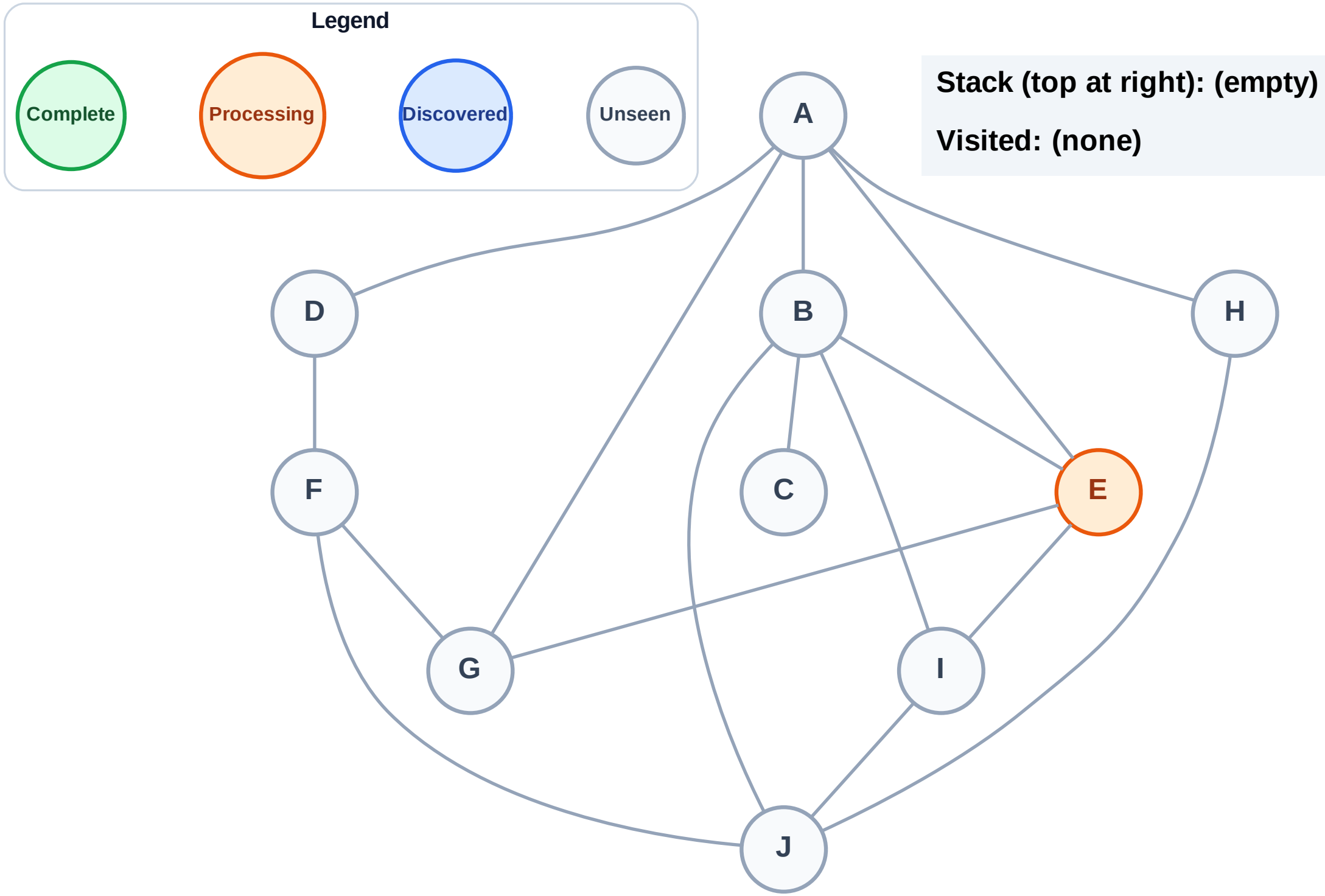
DFS Traversal

Step 1: Start at E



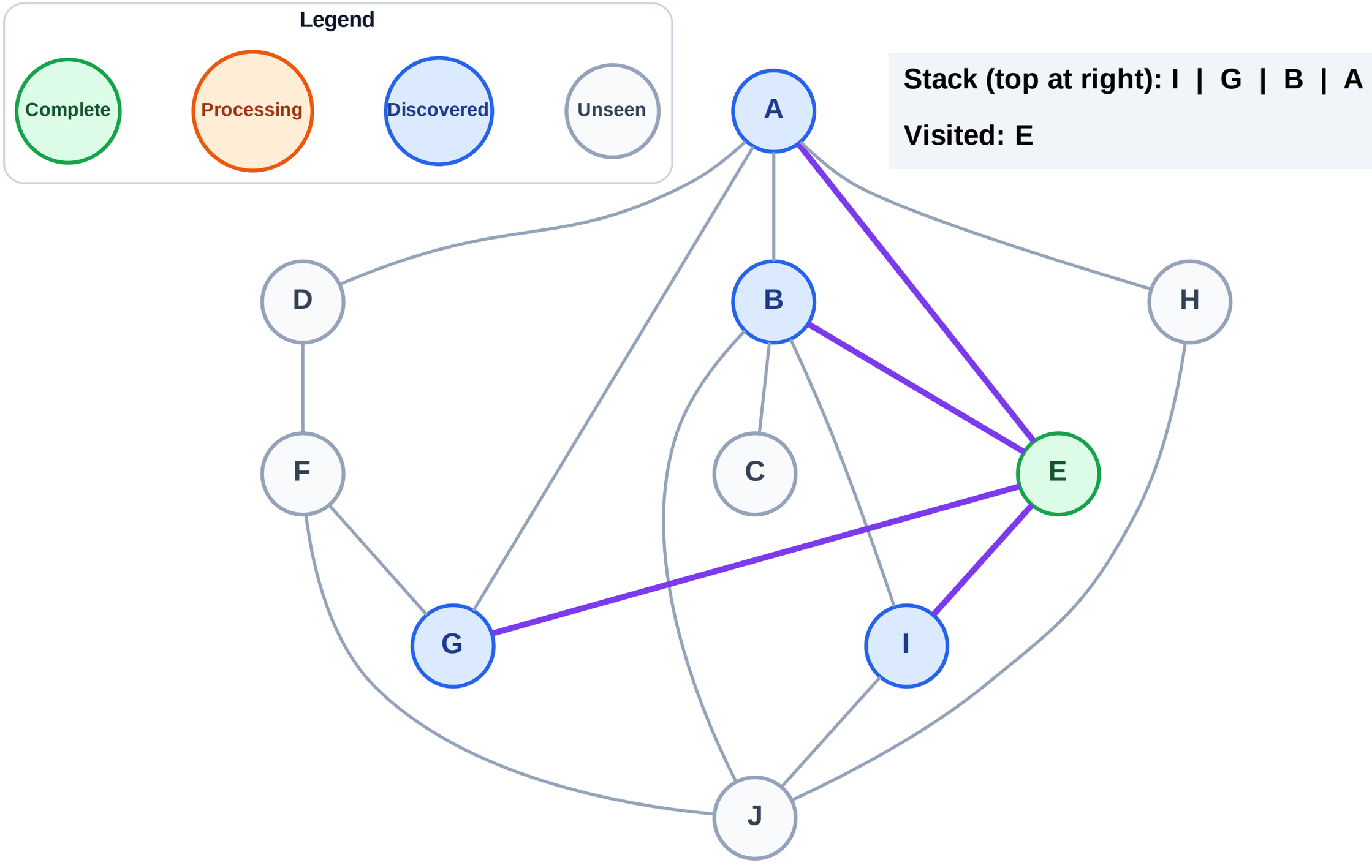
DFS Traversal

Step 2: Process node E



DFS Traversal

Step 3: Discover I, G, B, A from E

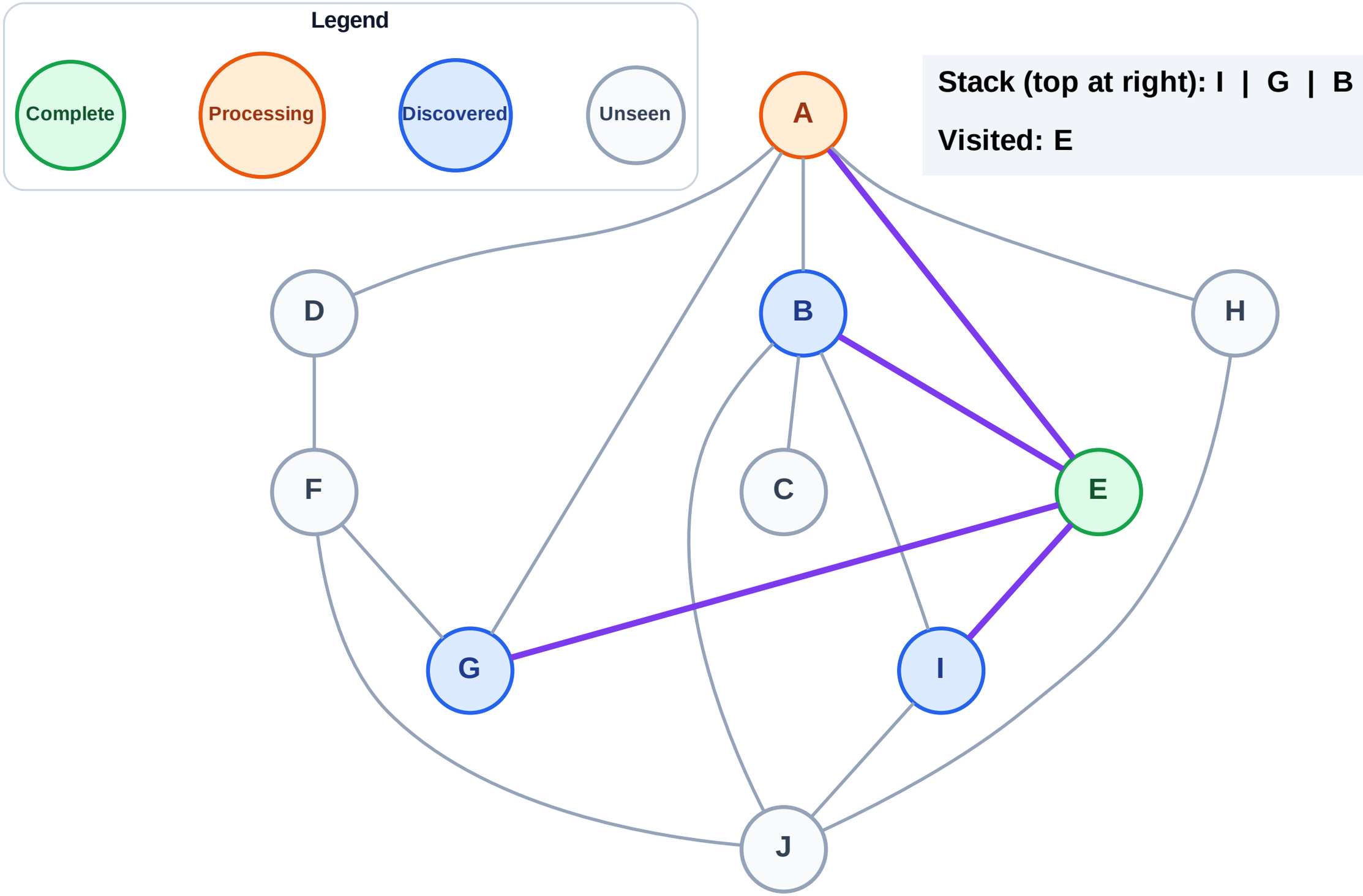


Stack (top at right): I | G | B | A

Visited: E

DFS Traversal

Step 4: Process node A

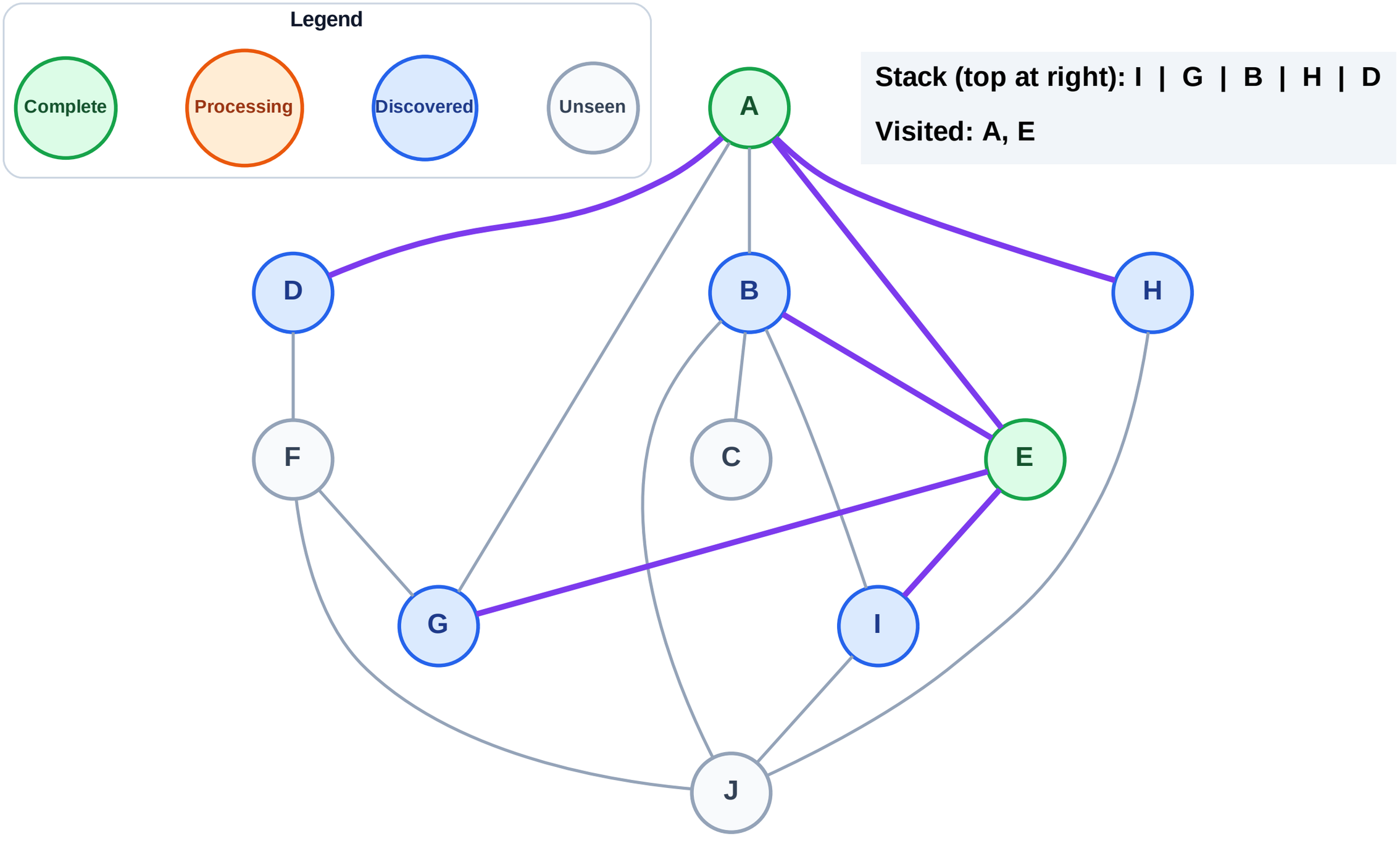


Stack (top at right): I | G | B

Visited: E

DFS Traversal

Step 5: Discover H, D from A

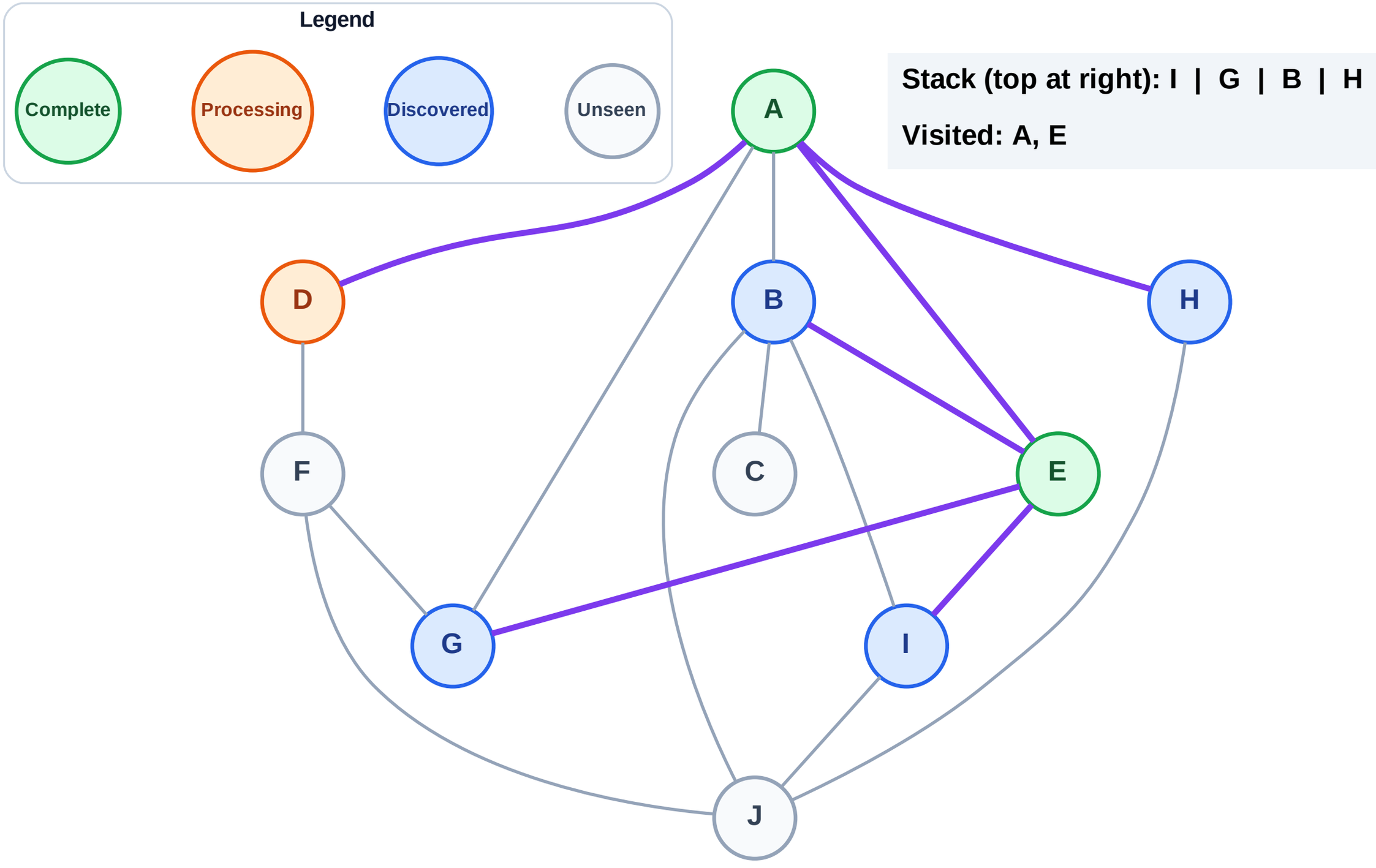


Stack (top at right): I | G | B | H | D

Visited: A, E

DFS Traversal

Step 6: Process node D

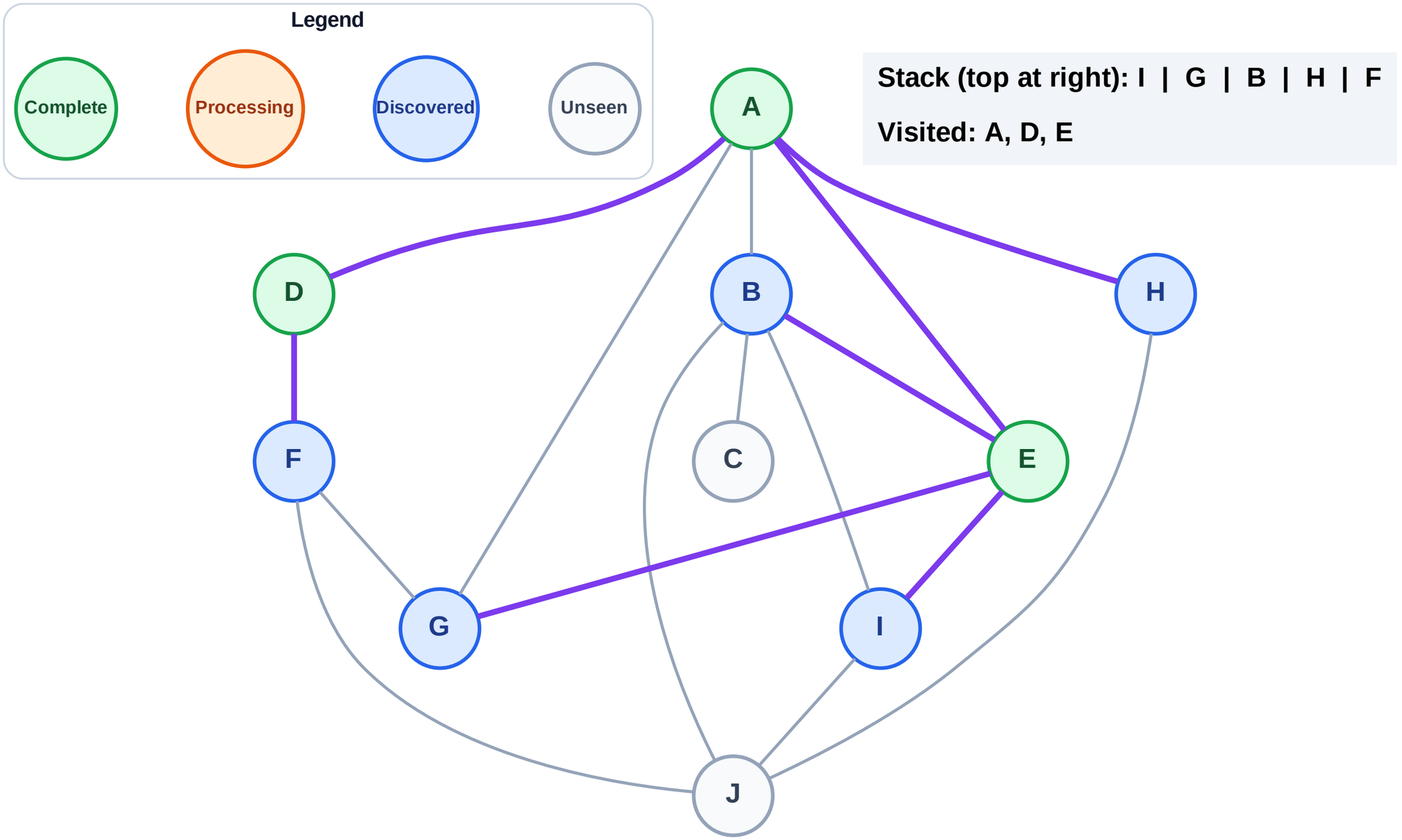


Stack (top at right): I | G | B | H

Visited: A, E

DFS Traversal

Step 7: Discover F from D

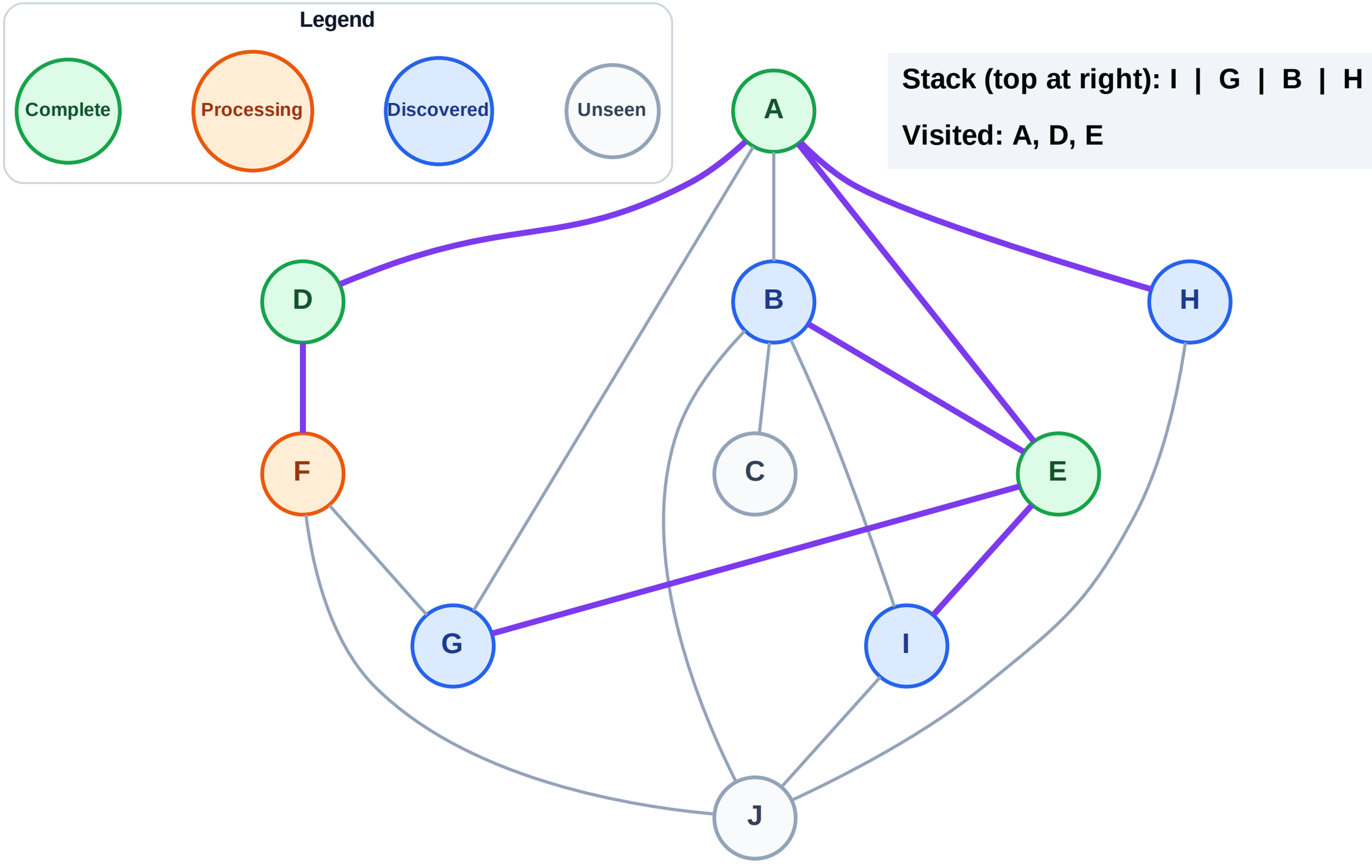


Stack (top at right): I | G | B | H | F

Visited: A, D, E

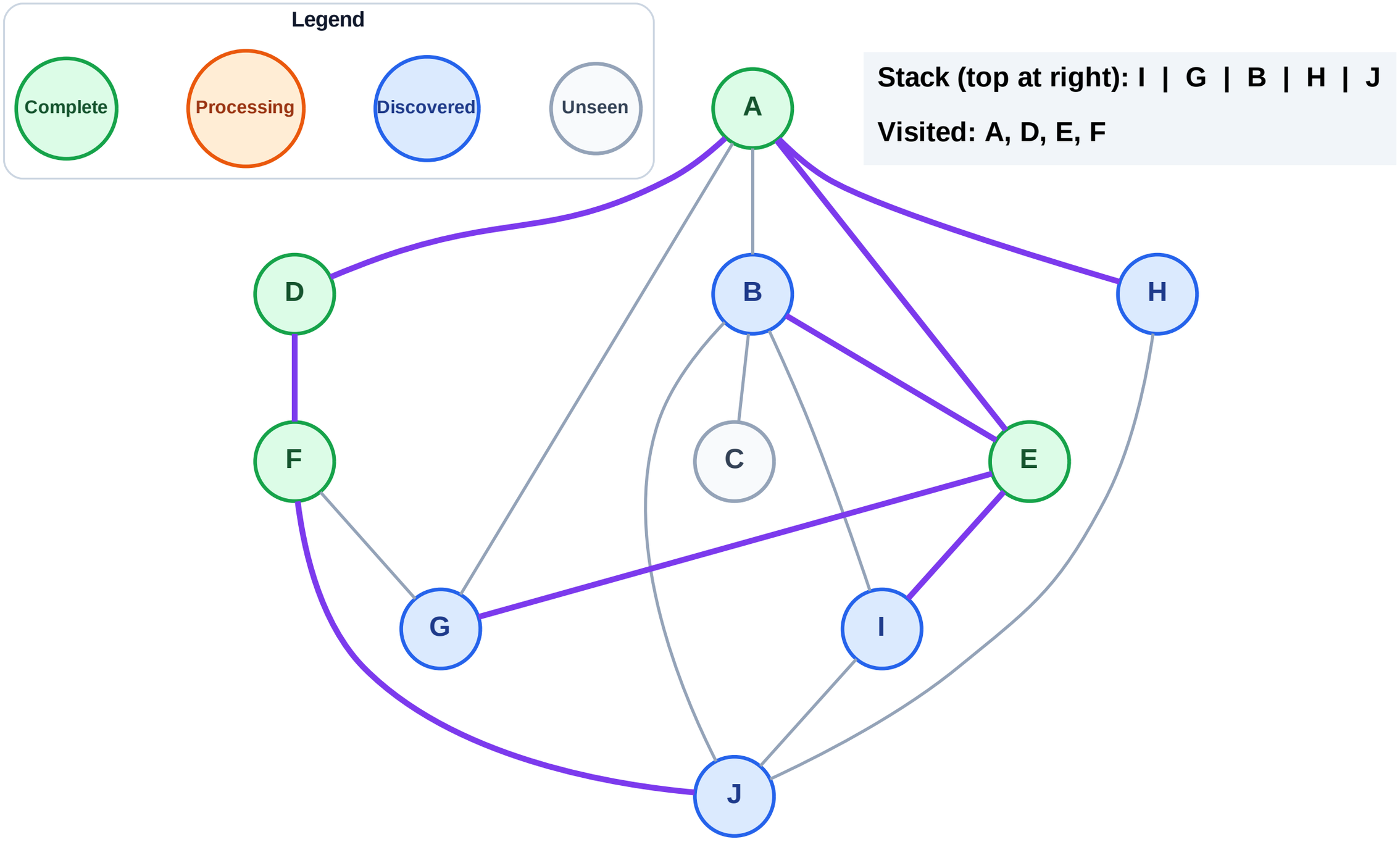
DFS Traversal

Step 8: Process node F



DFS Traversal

Step 9: Discover J from F

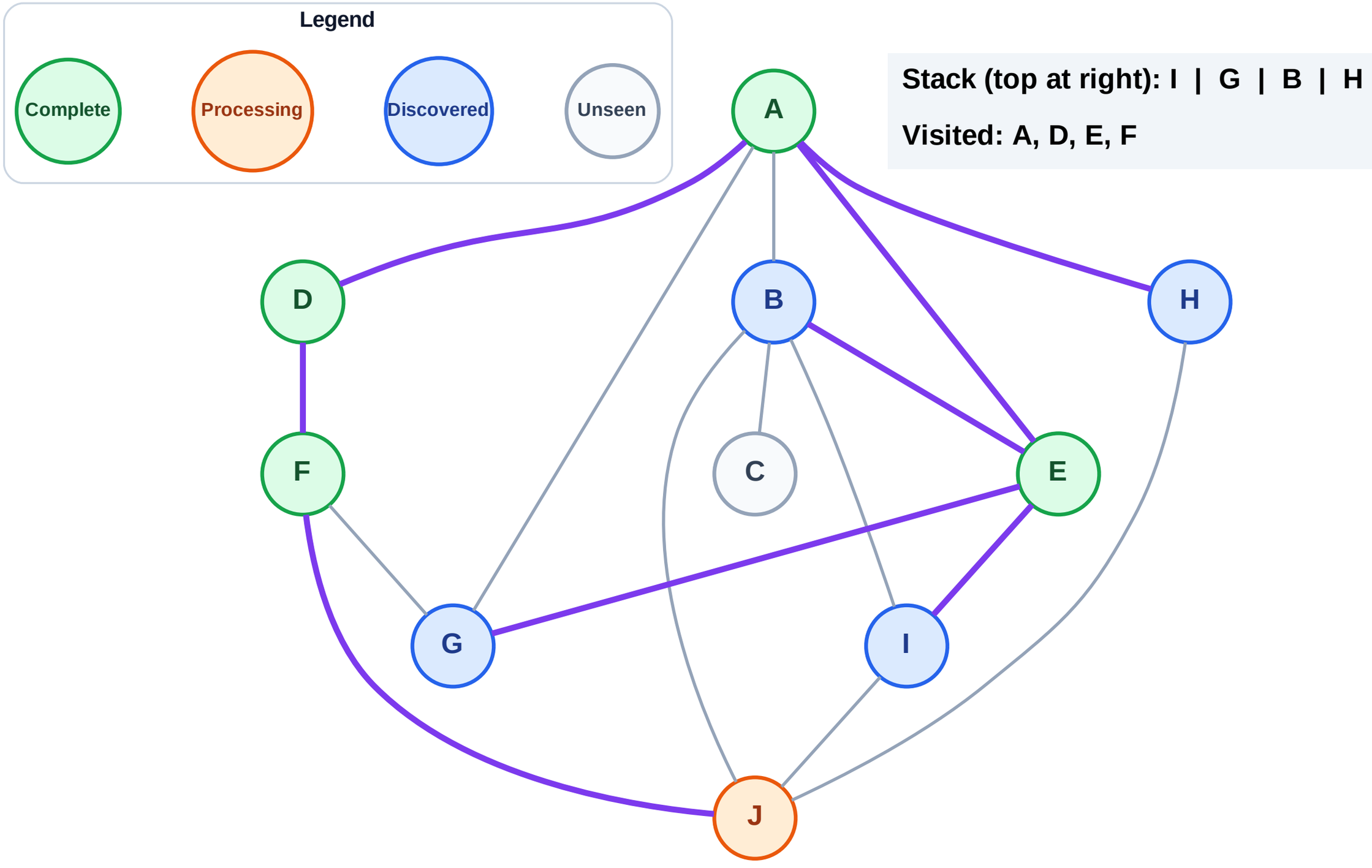


Stack (top at right): I | G | B | H | J

Visited: A, D, E, F

DFS Traversal

Step 10: Process node J

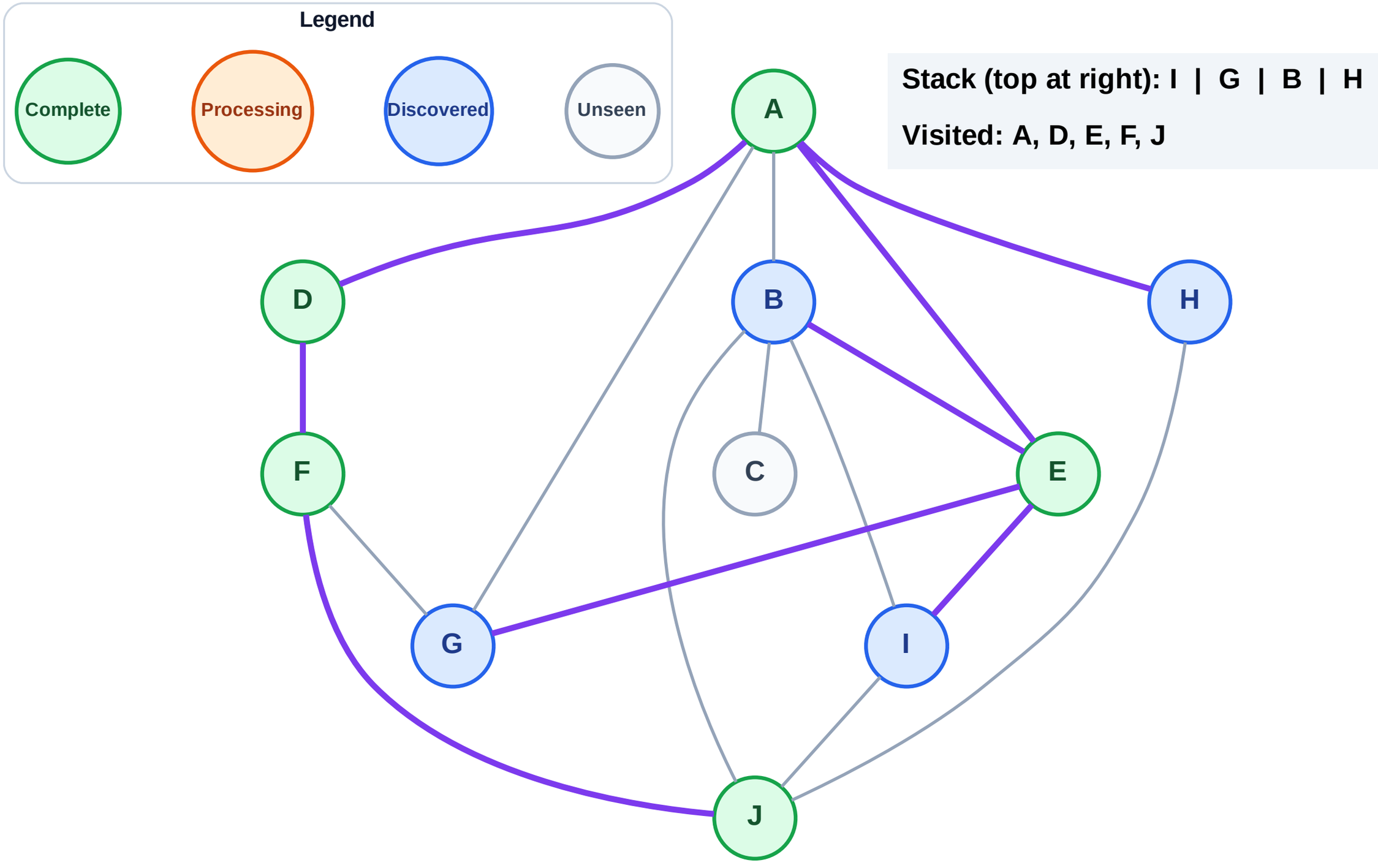


Stack (top at right): I | G | B | H

Visited: A, D, E, F

DFS Traversal

Step 11: No new neighbors from J

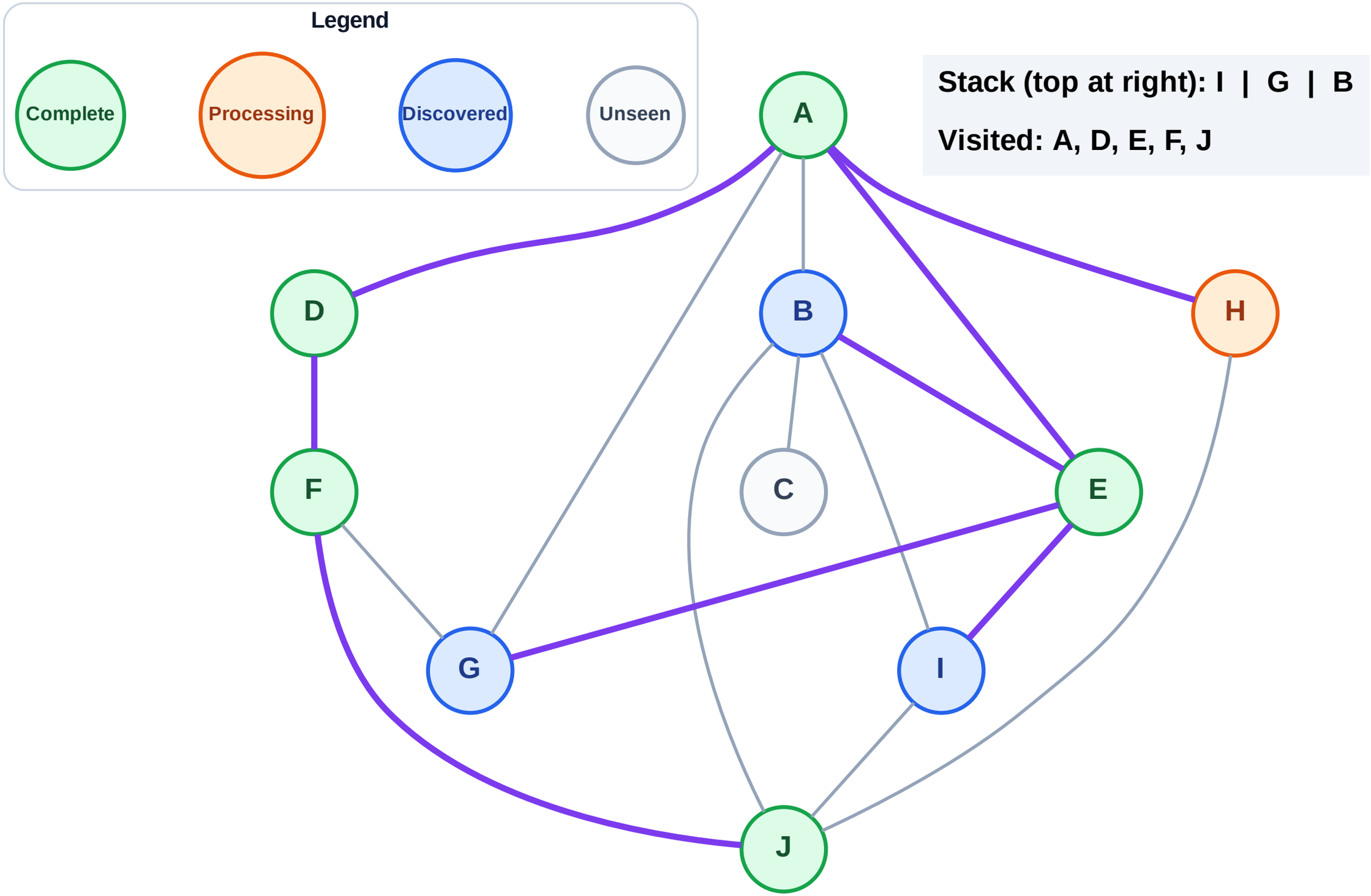


Stack (top at right): I | G | B | H

Visited: A, D, E, F, J

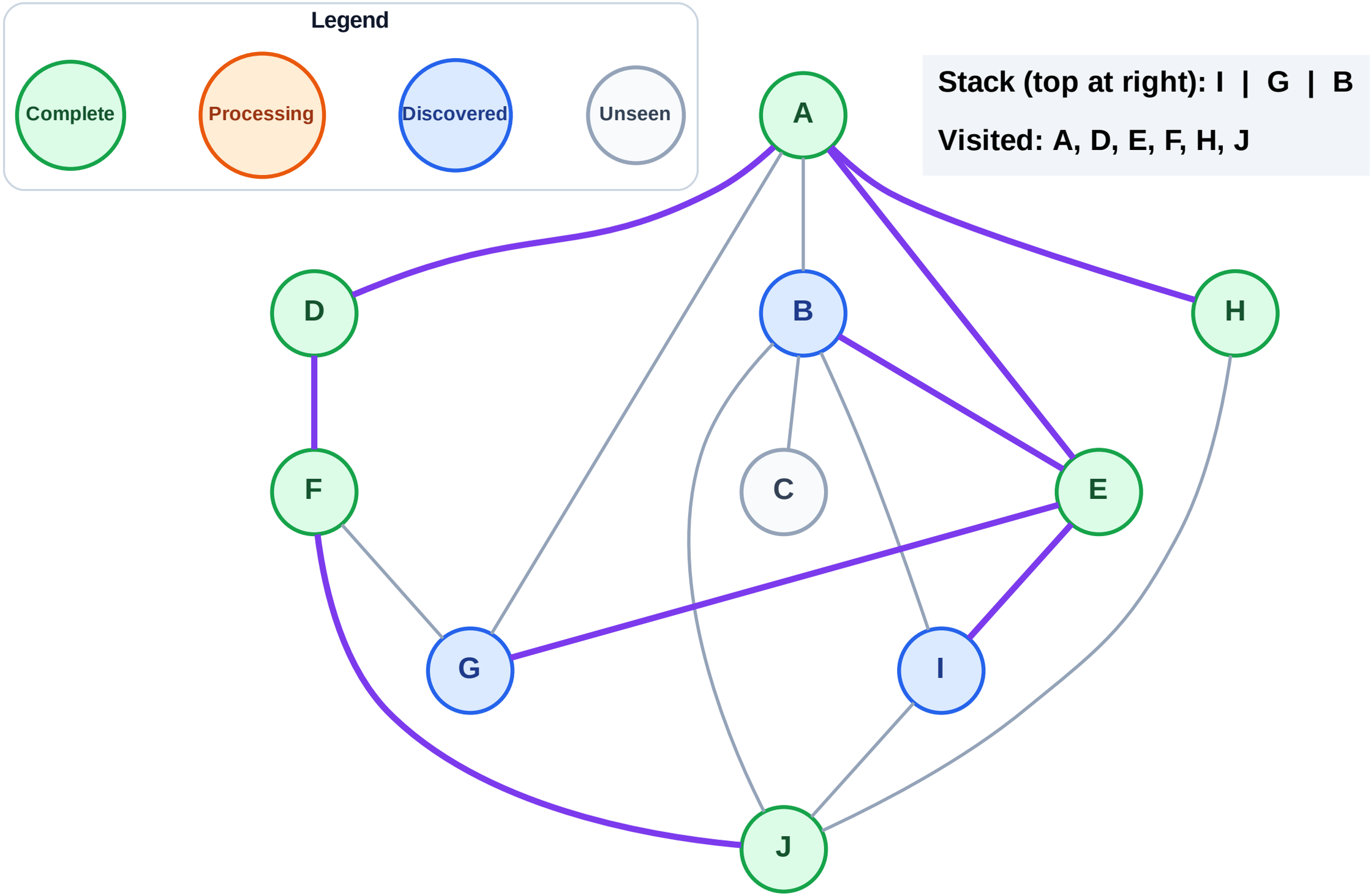
DFS Traversal

Step 12: Process node H



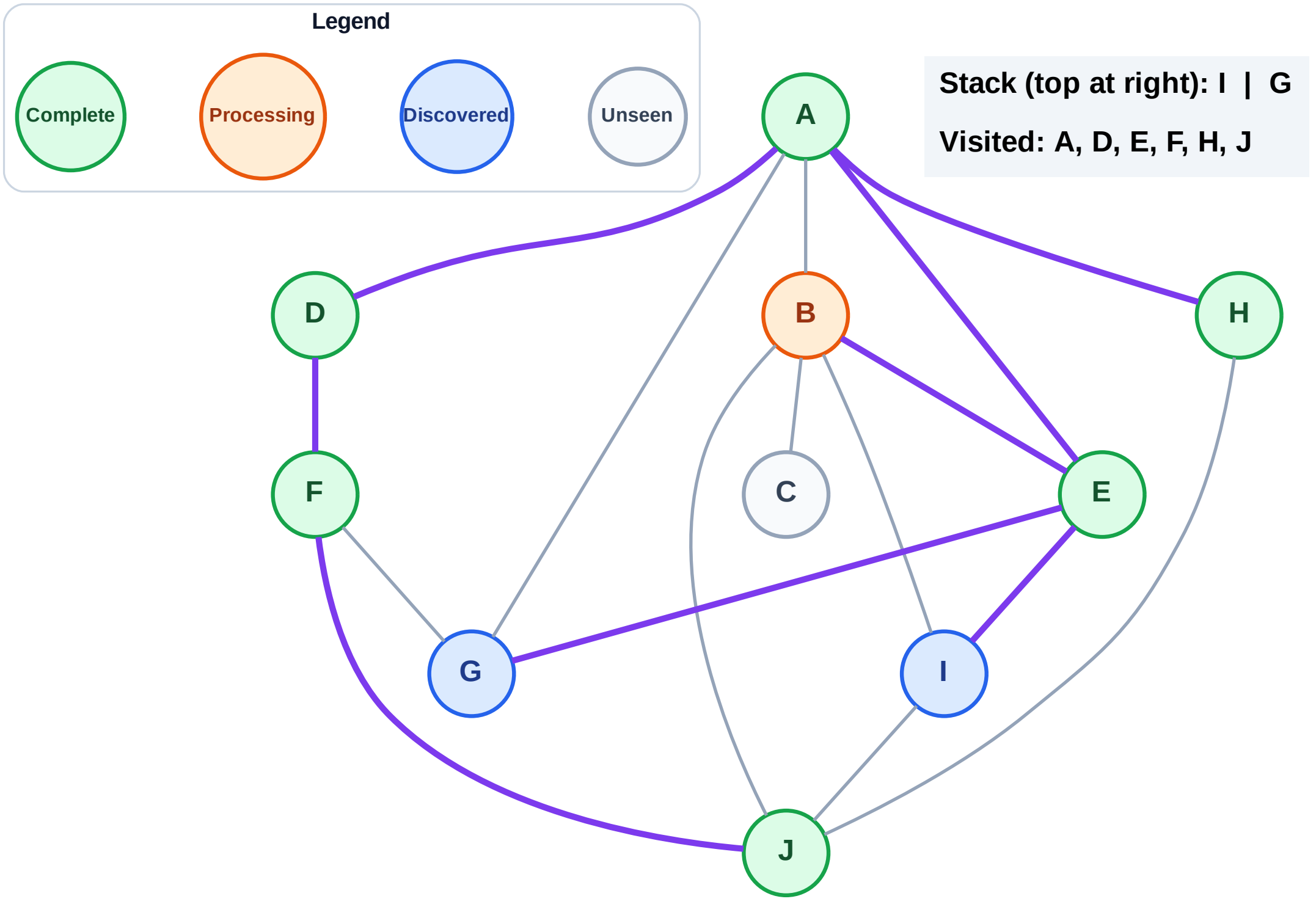
DFS Traversal

Step 13: No new neighbors from H



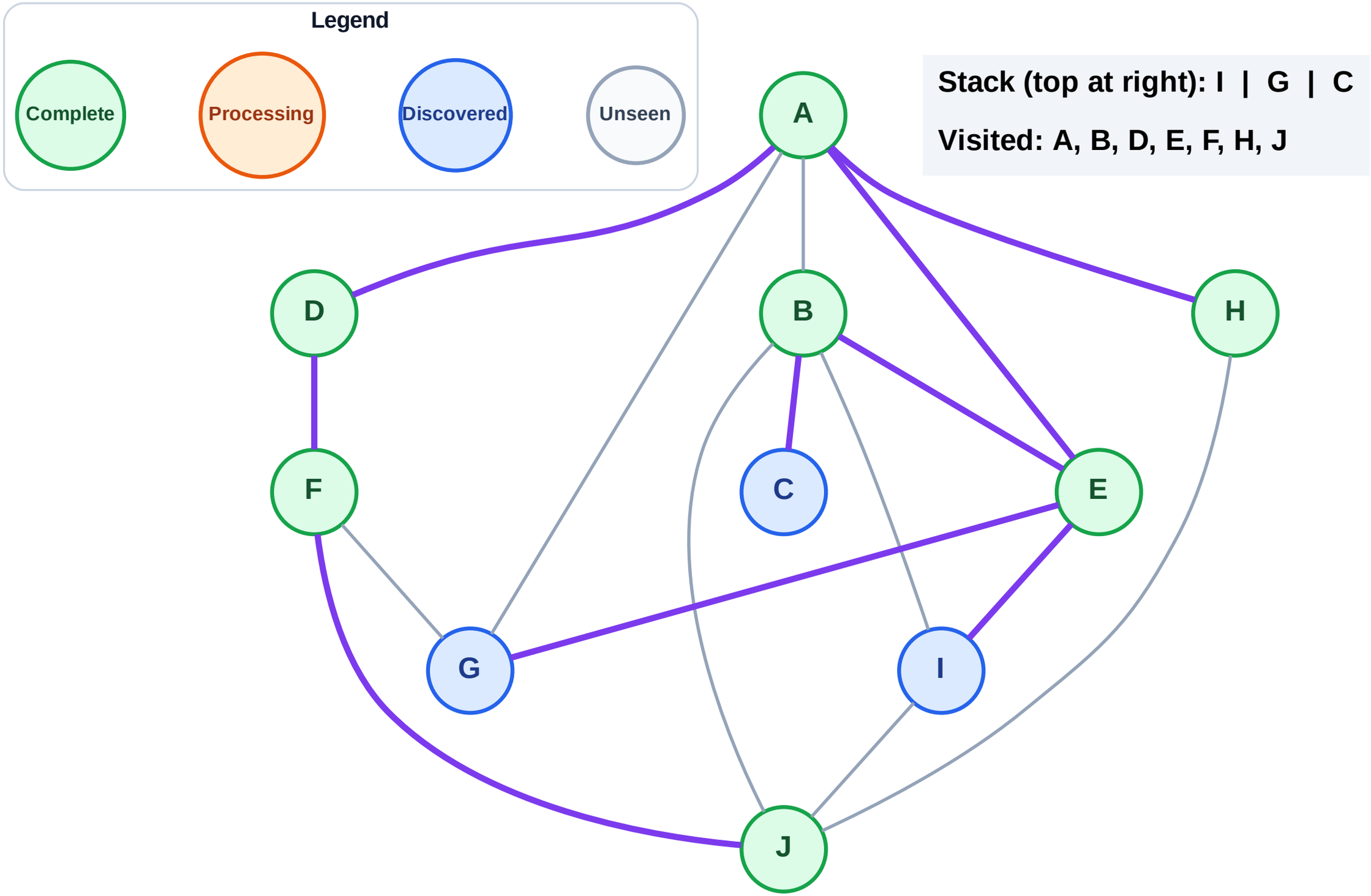
DFS Traversal

Step 14: Process node B



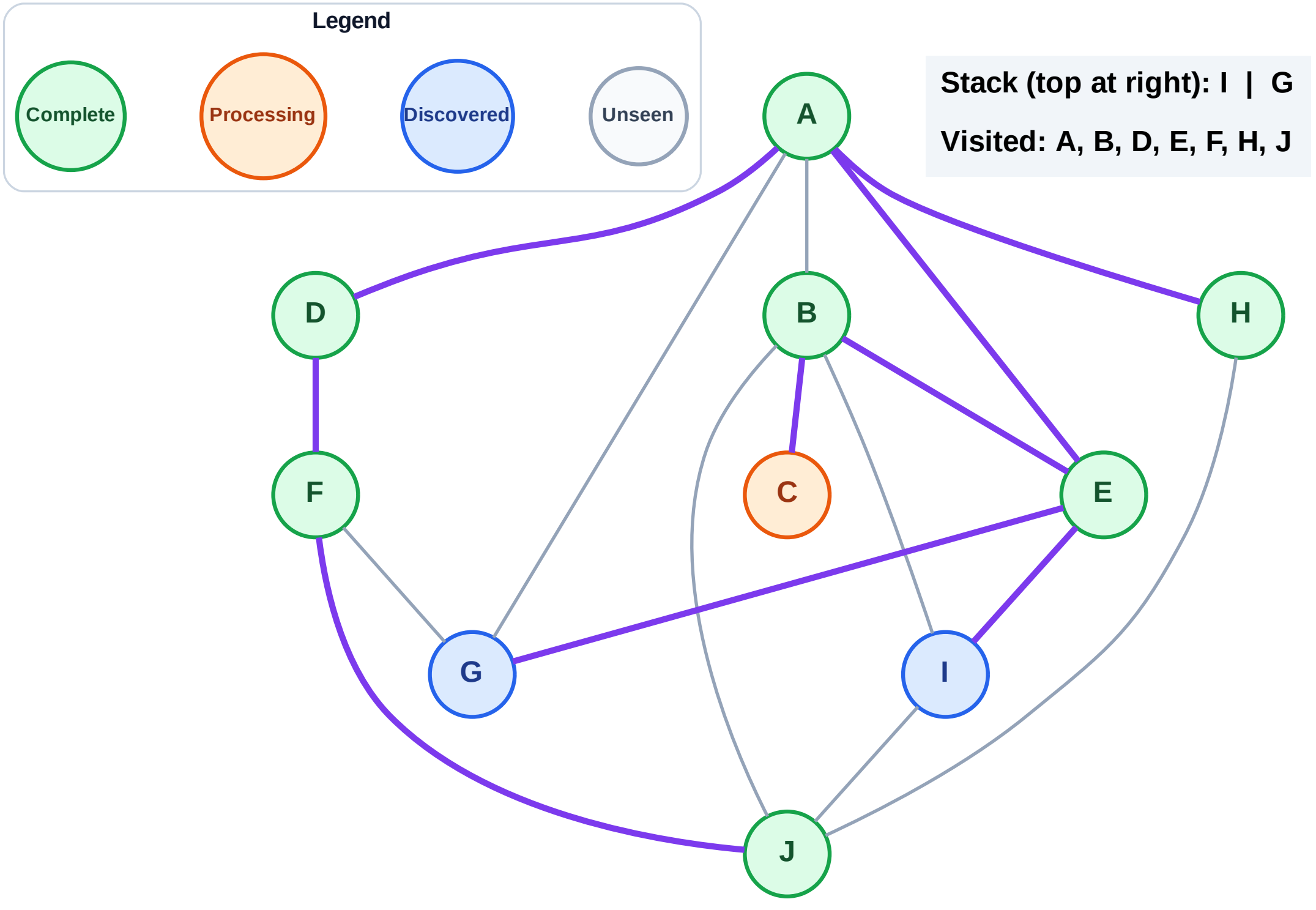
DFS Traversal

Step 15: Discover C from B



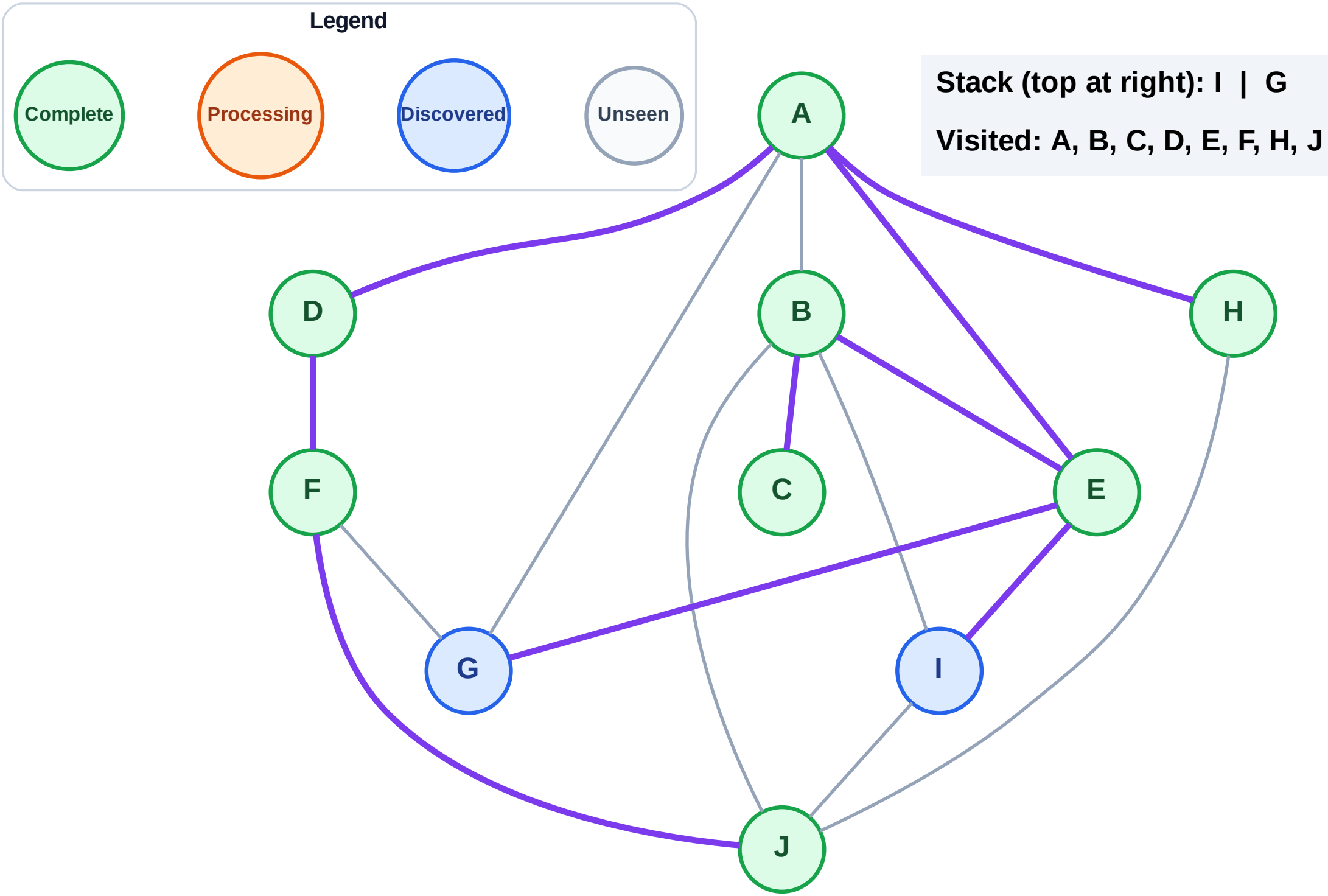
DFS Traversal

Step 16: Process node C



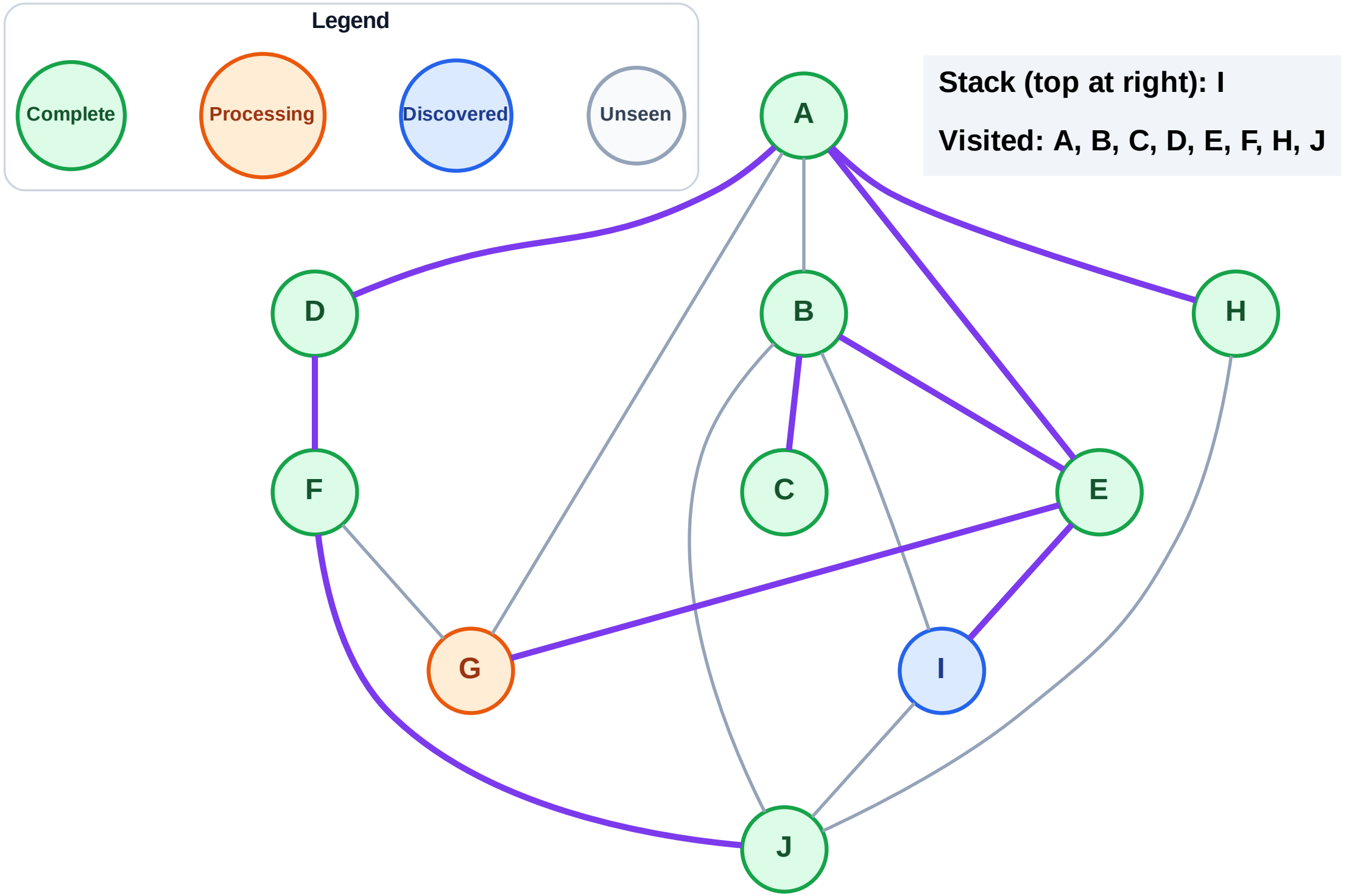
DFS Traversal

Step 17: No new neighbors from C



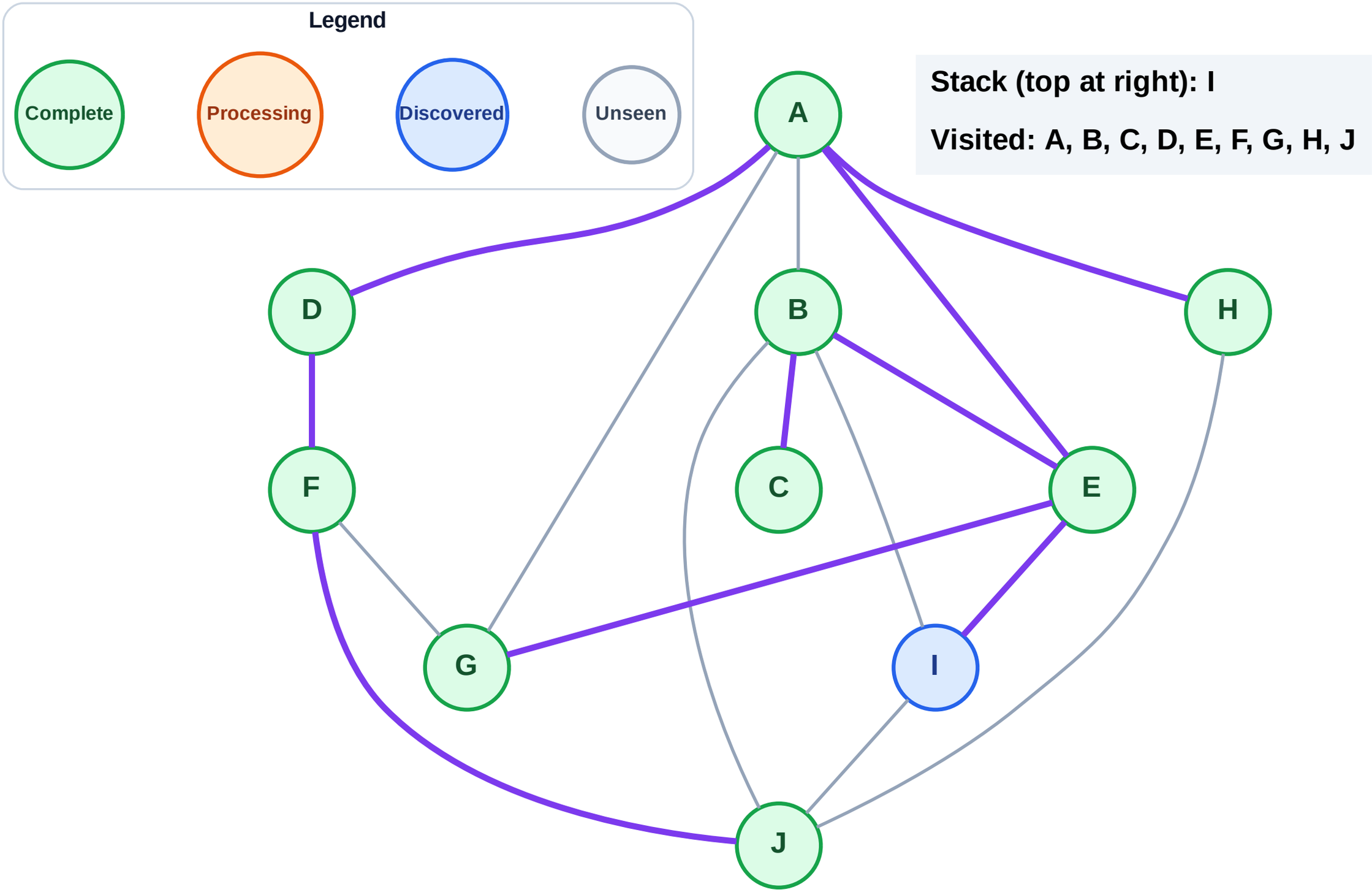
DFS Traversal

Step 18: Process node G



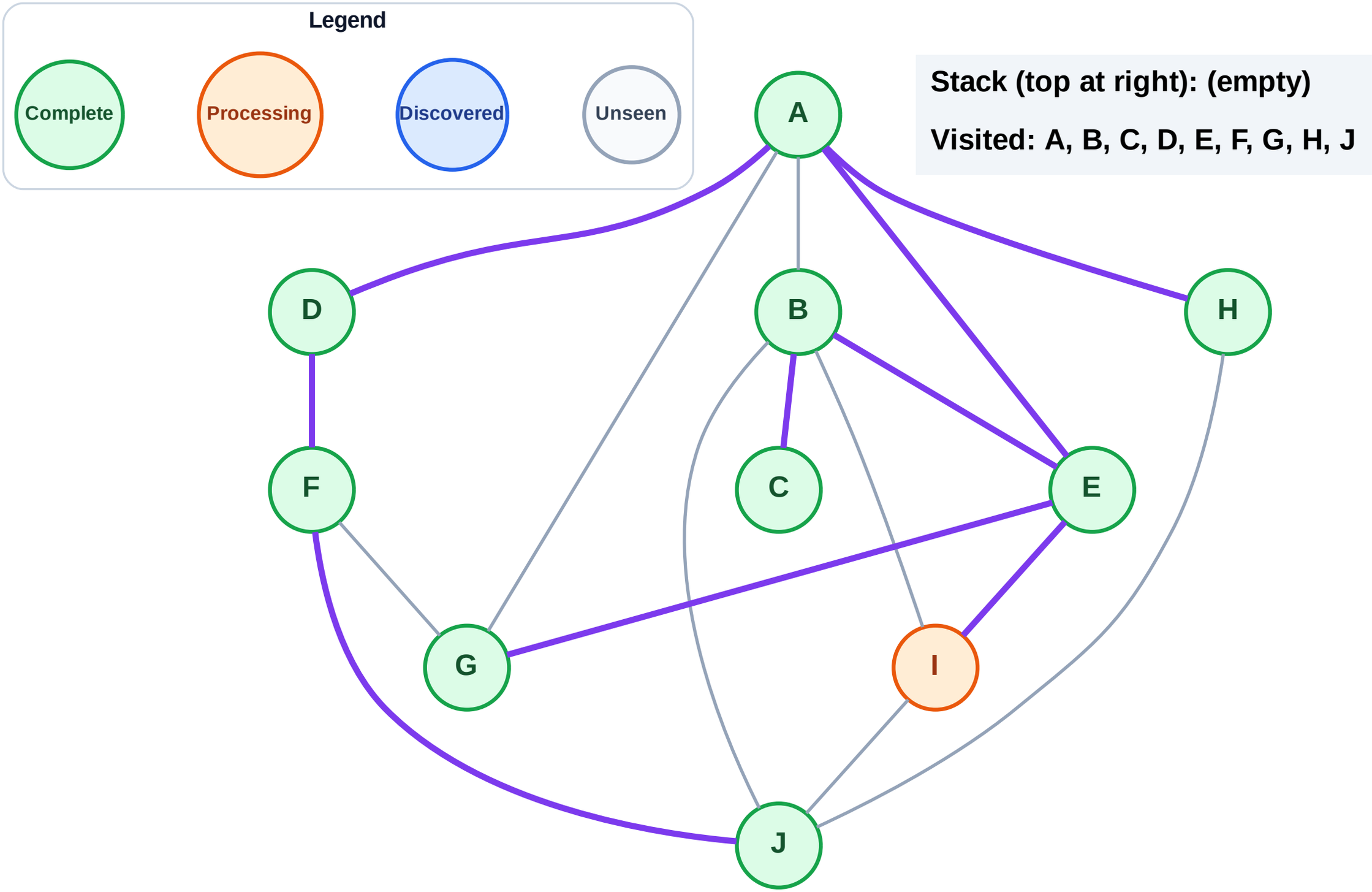
DFS Traversal

Step 19: No new neighbors from G



DFS Traversal

Step 20: Process node I



DFS Traversal

Step 21: No new neighbors from I

